US009715499B2

(12) United States Patent
Rubinger et al.

(10) Patent No.: US 9,715,499 B2
(45) Date of Patent: *Jul. 25, 2017

(54) **INTERACTIVE PATENT MAP**

(71) Applicant: **Global Prior Art, Inc.**, Boston, MA (US)

(72) Inventors: **Bruce Rubinger**, Newton, MA (US); **Vinh Do**, West Roxbury, MA (US); **Alexander Kaufman**, Chestnut Hill, MA (US); **Kuppar Ranganath Udupa**, Ottawa (CA)

(73) Assignee: **GLOBAL PRIOR ART, INC.**, Boston, MA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 1161 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **13/676,078**

(22) Filed: **Nov. 13, 2012**

(65) **Prior Publication Data**

US 2013/0159308 A1 Jun. 20, 2013

Related U.S. Application Data

(63) Continuation of application No. 12/587,666, filed on Oct. 8, 2009, now Pat. No. 8,335,998, which is a

(Continued)

(51) **Int. Cl.**
G06F 3/0481 (2013.01)
G06F 17/30 (2006.01)

(Continued)

(52) **U.S. Cl.**
CPC ***G06F 17/30011*** (2013.01); ***G06F 3/0481*** (2013.01); ***G06F 3/0482*** (2013.01);

(Continued)

(58) **Field of Classification Search**
CPC G06F 3/0481; G06F 3/0482; G06F 3/0484; G06F 3/0486; H04N 41/12
See application file for complete search history.

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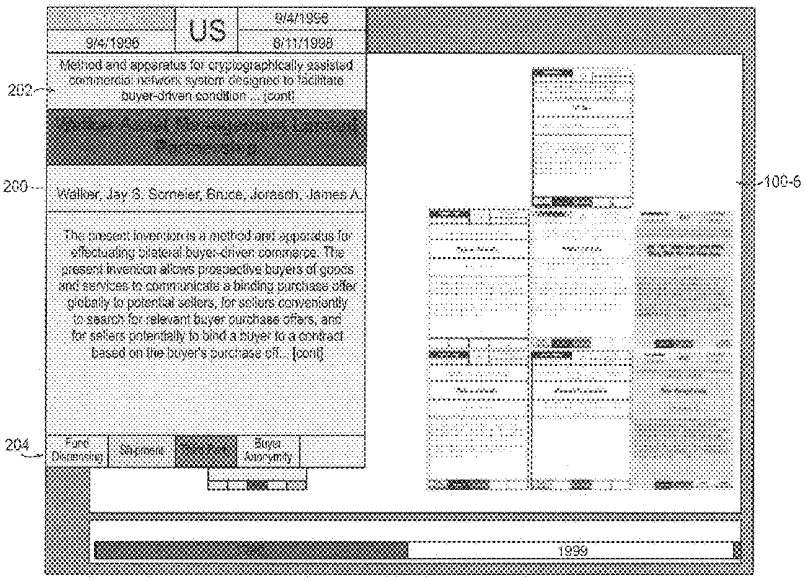
Primary Examiner — Steven Sax
(74) *Attorney, Agent, or Firm* — Hamilton, Brook, Smith & Reynolds, P.C.

(57) **ABSTRACT**

Systems and methods are provided for generating an interactive map for displaying and analyzing a compressive intellectual property landscape within a given field. Based on content analysis of relevant patents and patent applications, this prior art map provides a systematic review of vast quantities of data, thereby allowing the user to discern critical technology and product trends, prior art references, and the strategies of both leading and emerging competitors. Each patent represented on the map can be analyzed within the context of the prior art landscape to uncover novel features, strong claims, and business and technology trends. This comprehensive view can provide a foundation for creating effective corporate strategies in-tune with the realities of the intellectual property terrain.

26 Claims, 37 Drawing Sheets
(30 of 37 Drawing Sheet(s) Filed in Color)



Related U.S. Application Data

continuation of application No. 11/648,034, filed on Dec. 29, 2006, now abandoned.

(51) **Int. Cl.**

H04L 12/24	(2006.01)
G06F 3/0482	(2013.01)
G06F 3/0484	(2013.01)
G06F 3/0486	(2013.01)
G06Q 50/18	(2012.01)

(52) **U.S. Cl.**

CPC ***G06F 3/0484*** (2013.01); ***G06F 3/0486*** (2013.01); ***G06Q 50/184*** (2013.01); ***H04L 41/12*** (2013.01)

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Legend
Branch 2
104
Branch 3
106
108

FIG. 1A

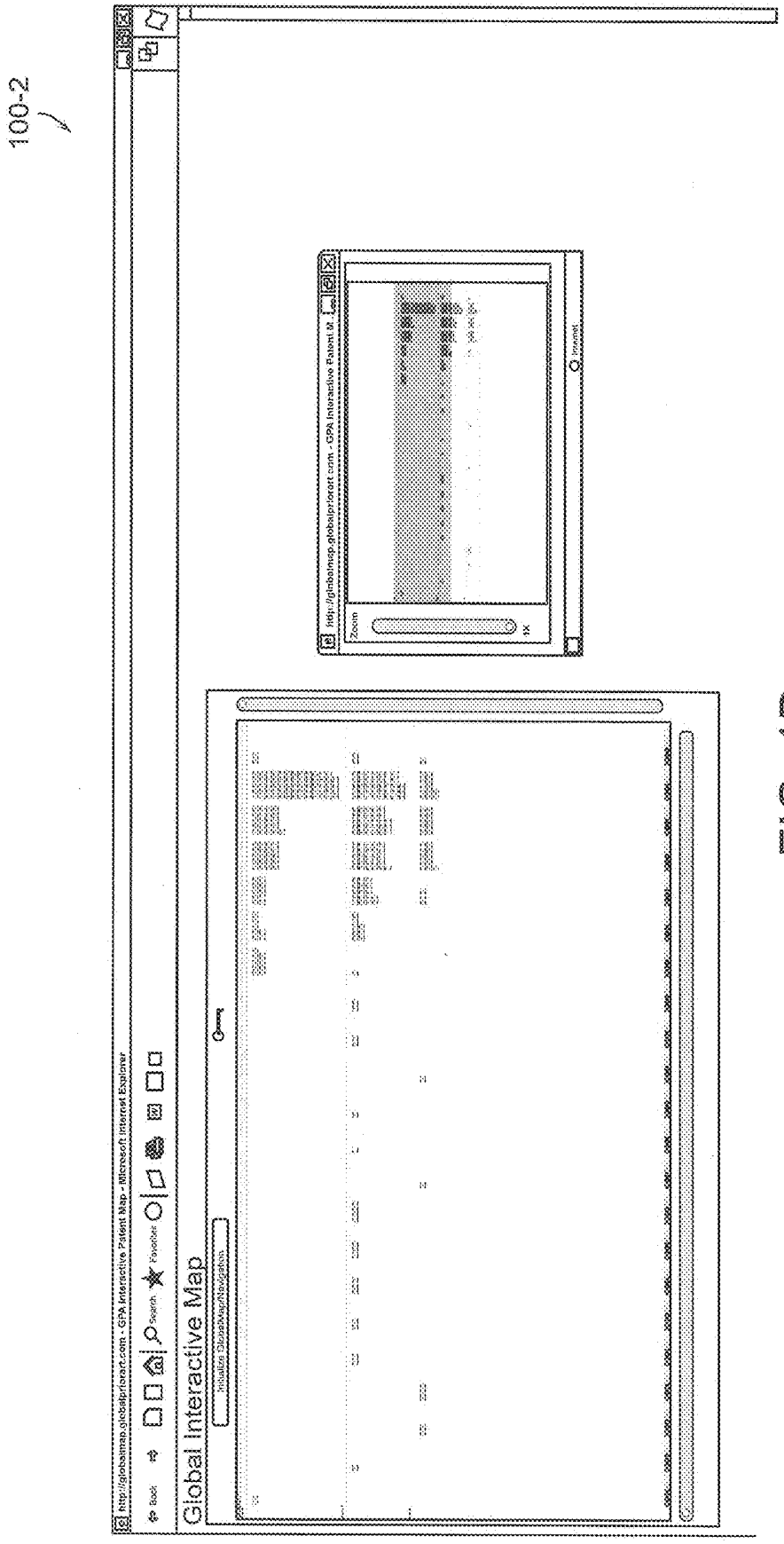
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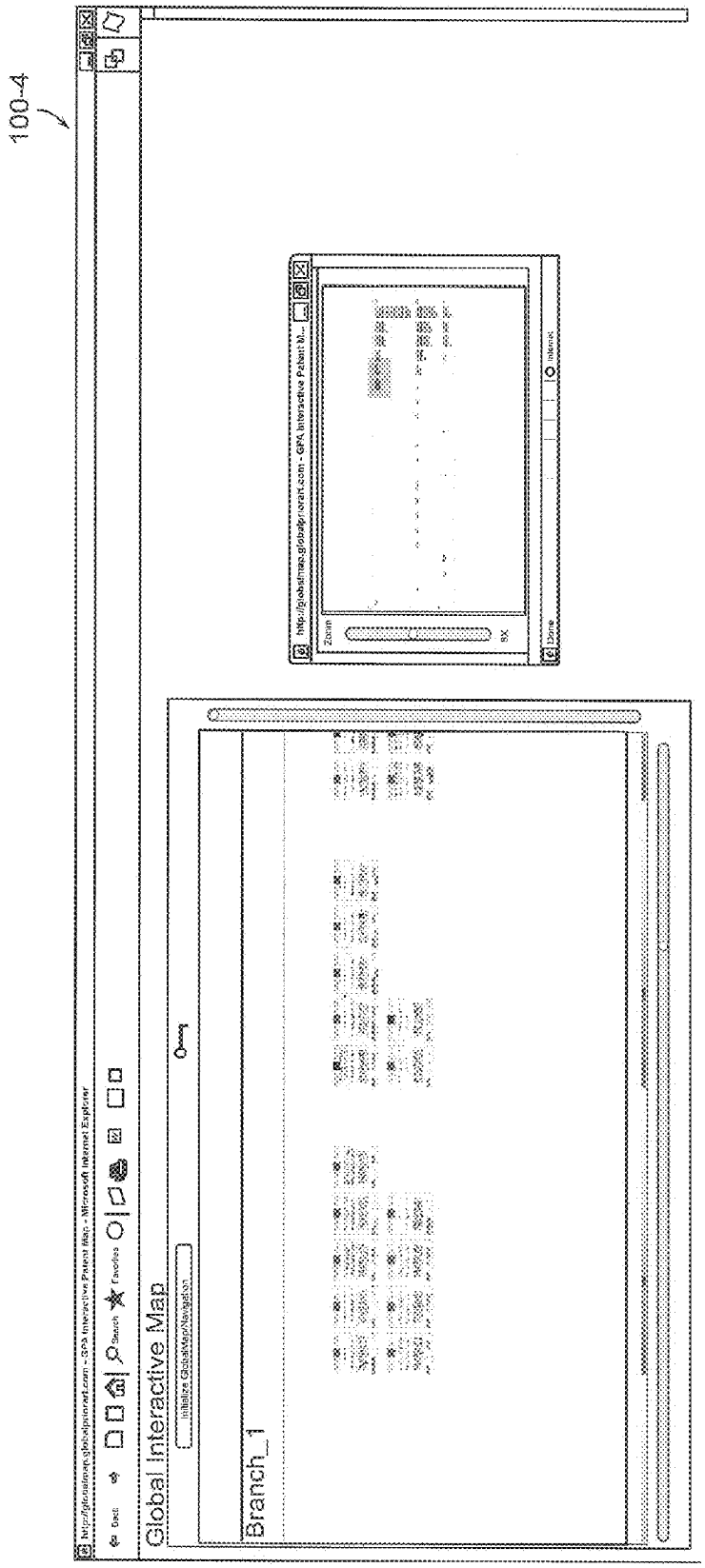
FIG. 1B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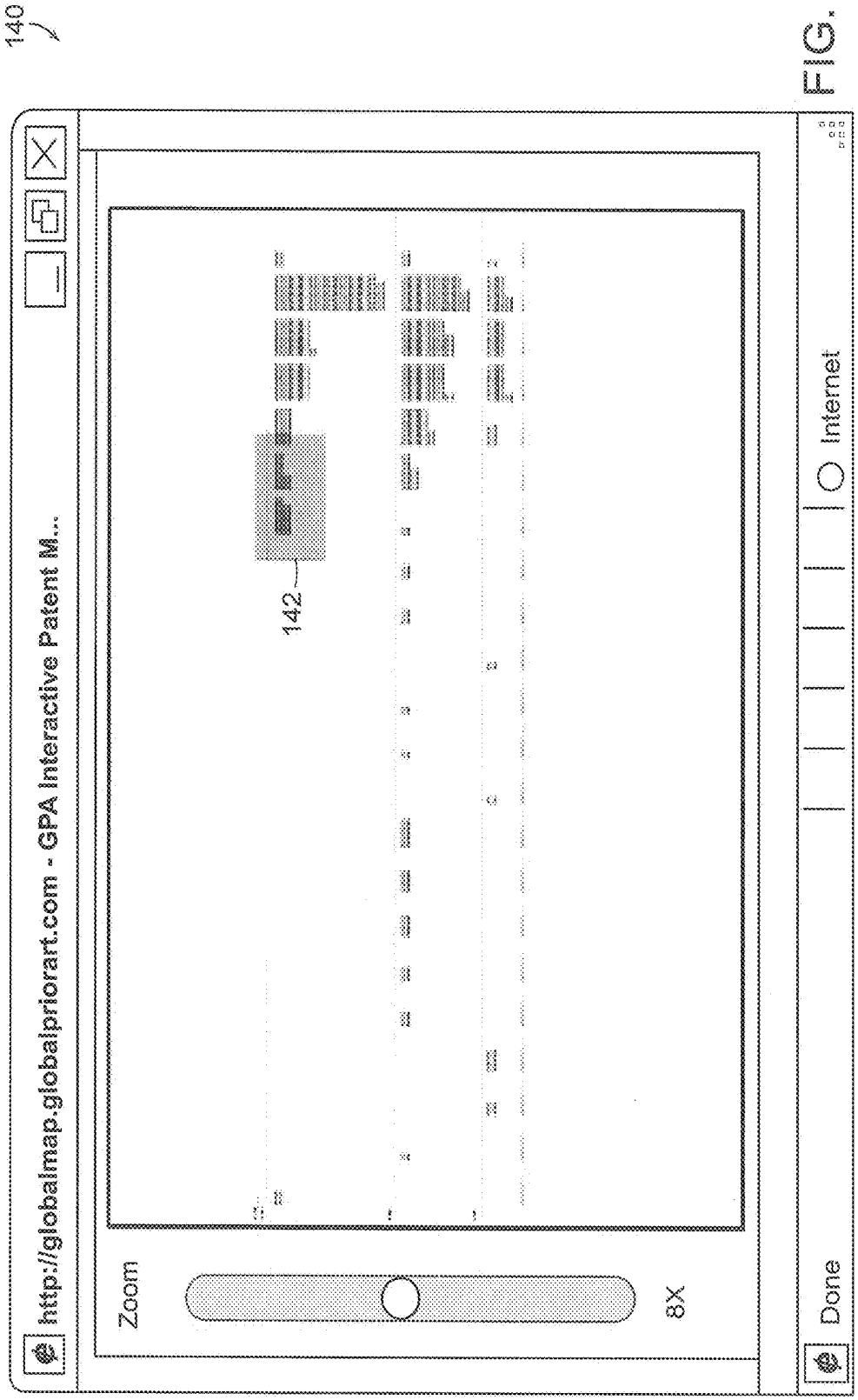
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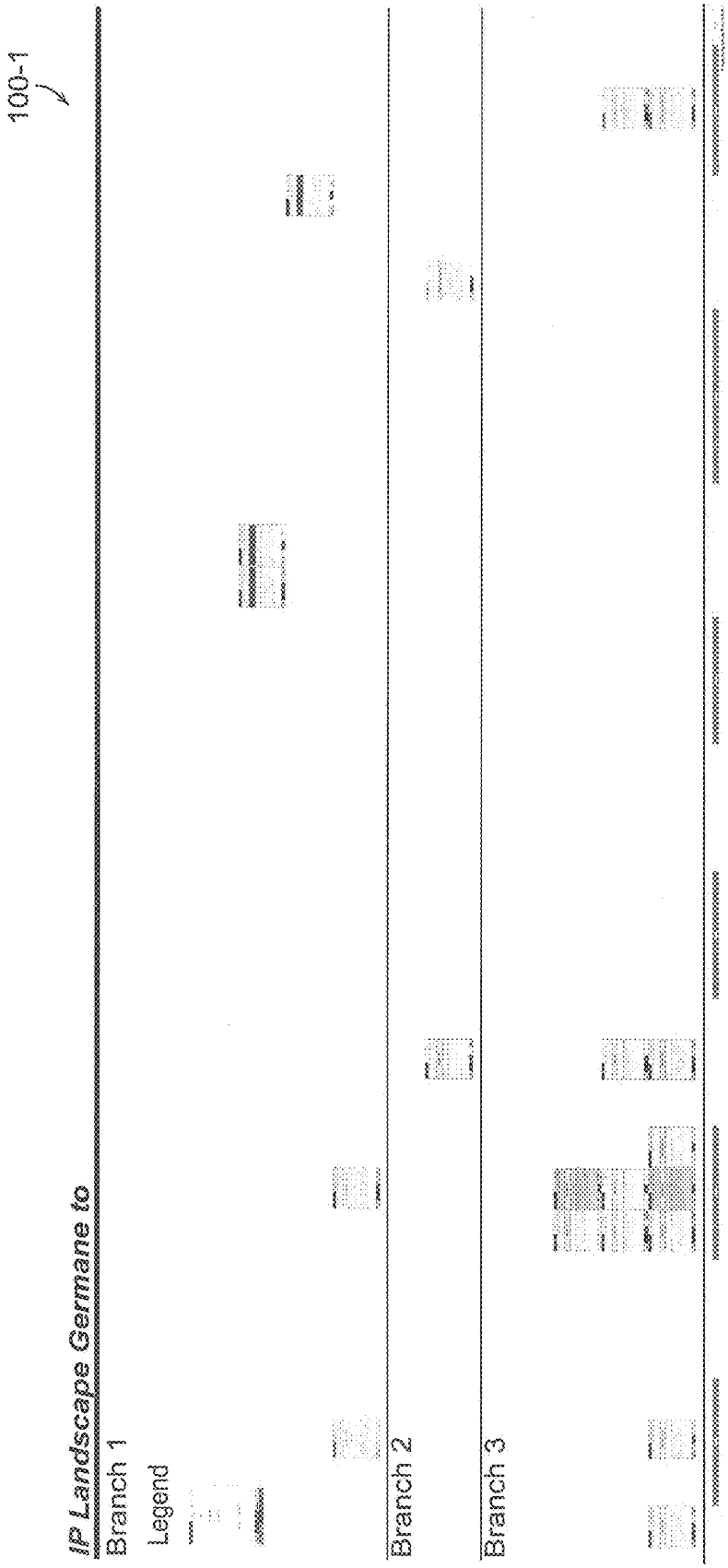
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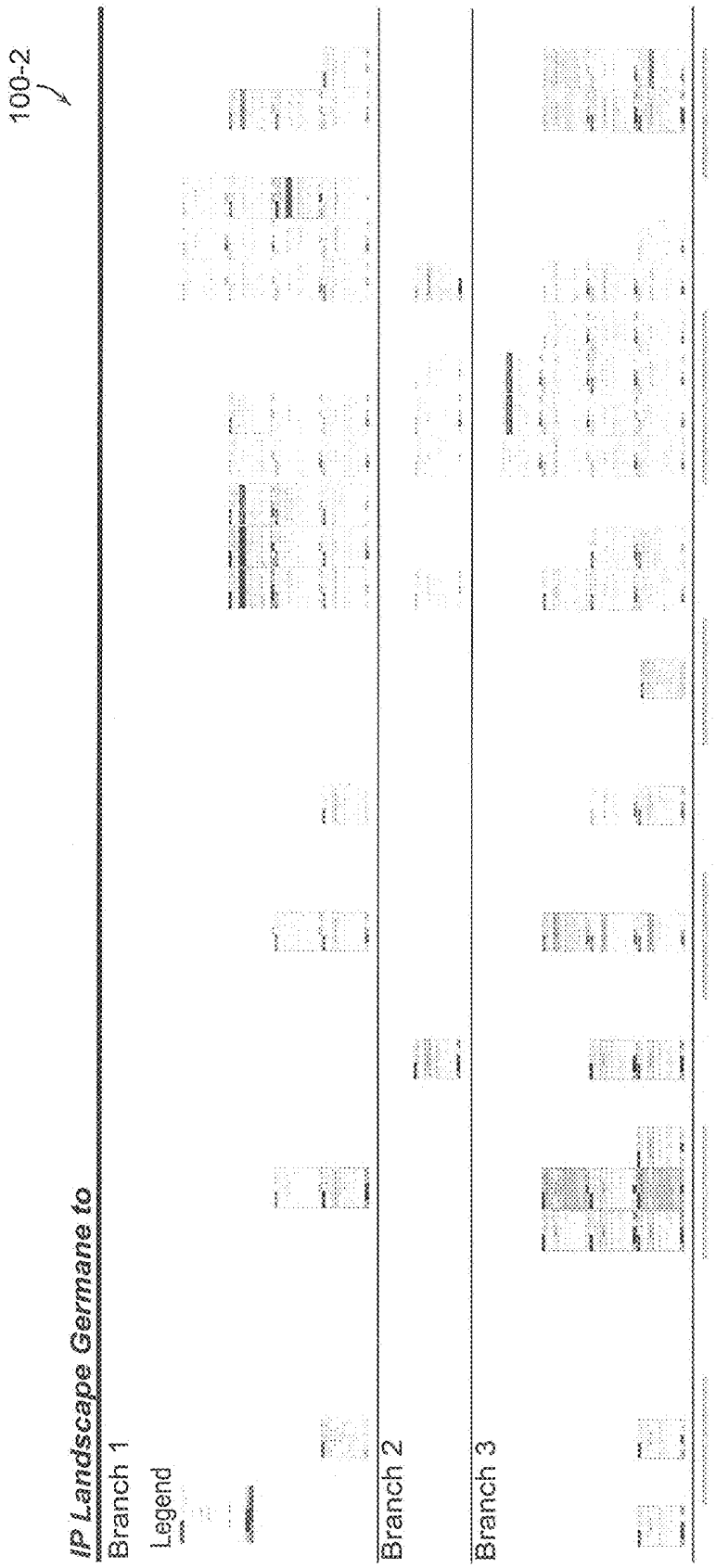
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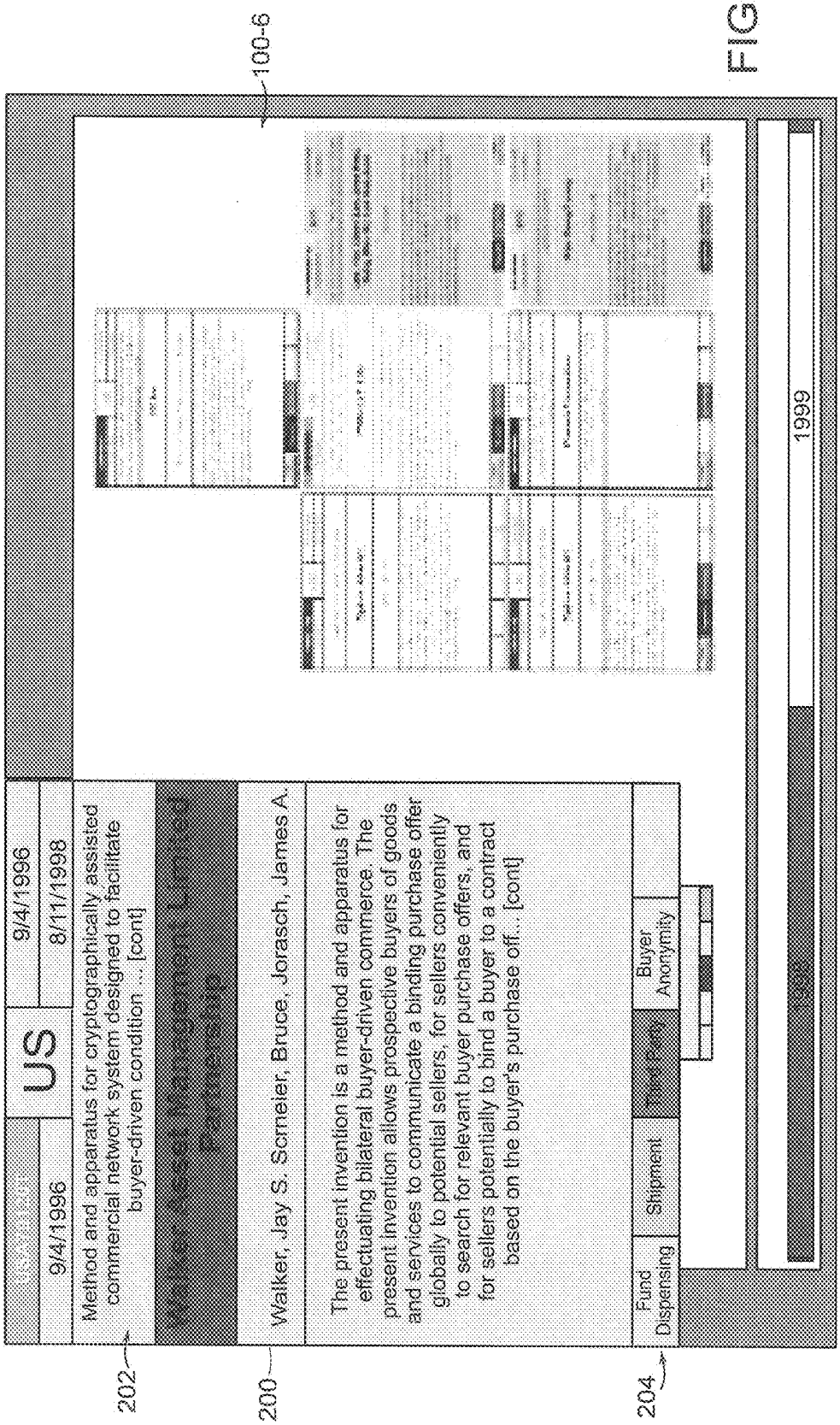
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US
9/4/1996
9/4/1996
8/11/1998
Method and apparatus for cryptographically assisted commercial network system designed to facilitate buyer-driven condition ... [cont]
Walker, Jay S. Scrneier, Bruce, Jorasch, James A.
The present invention is a method and apparatus for effectuating bilateral buyer-driven commerce. The present invention allows prospective buyers of goods and services to communicate a binding purchase offer globally to potential sellers, for sellers conveniently to search for relevant buyer purchase offers, and for sellers potentially to bind a buyer to a contract based on the buyer's purchase off... [cont]
Fund Dispensing
Shipment
Buyer Anonymity
1999
202
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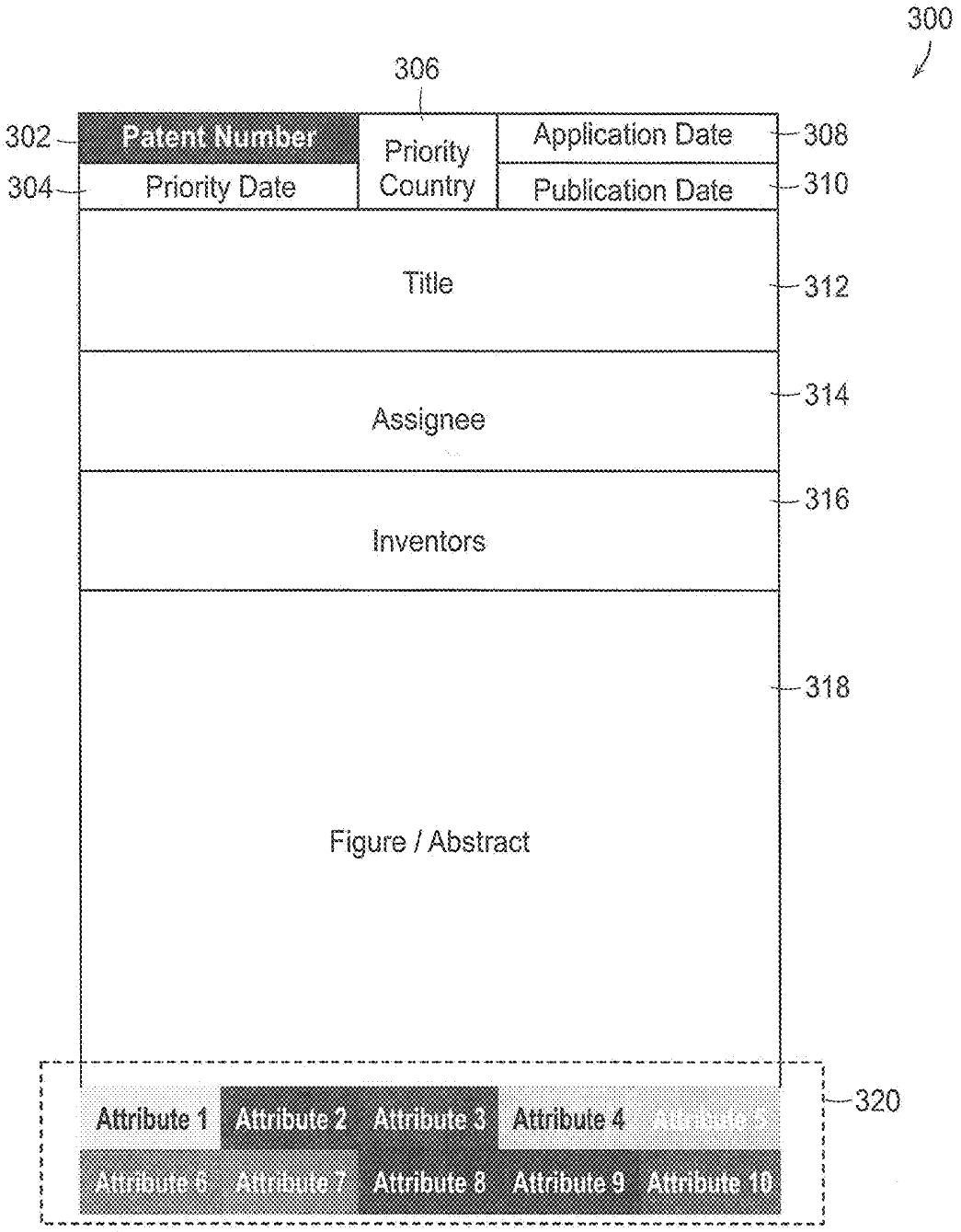
204

FIG. 2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Assignee
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Inventors
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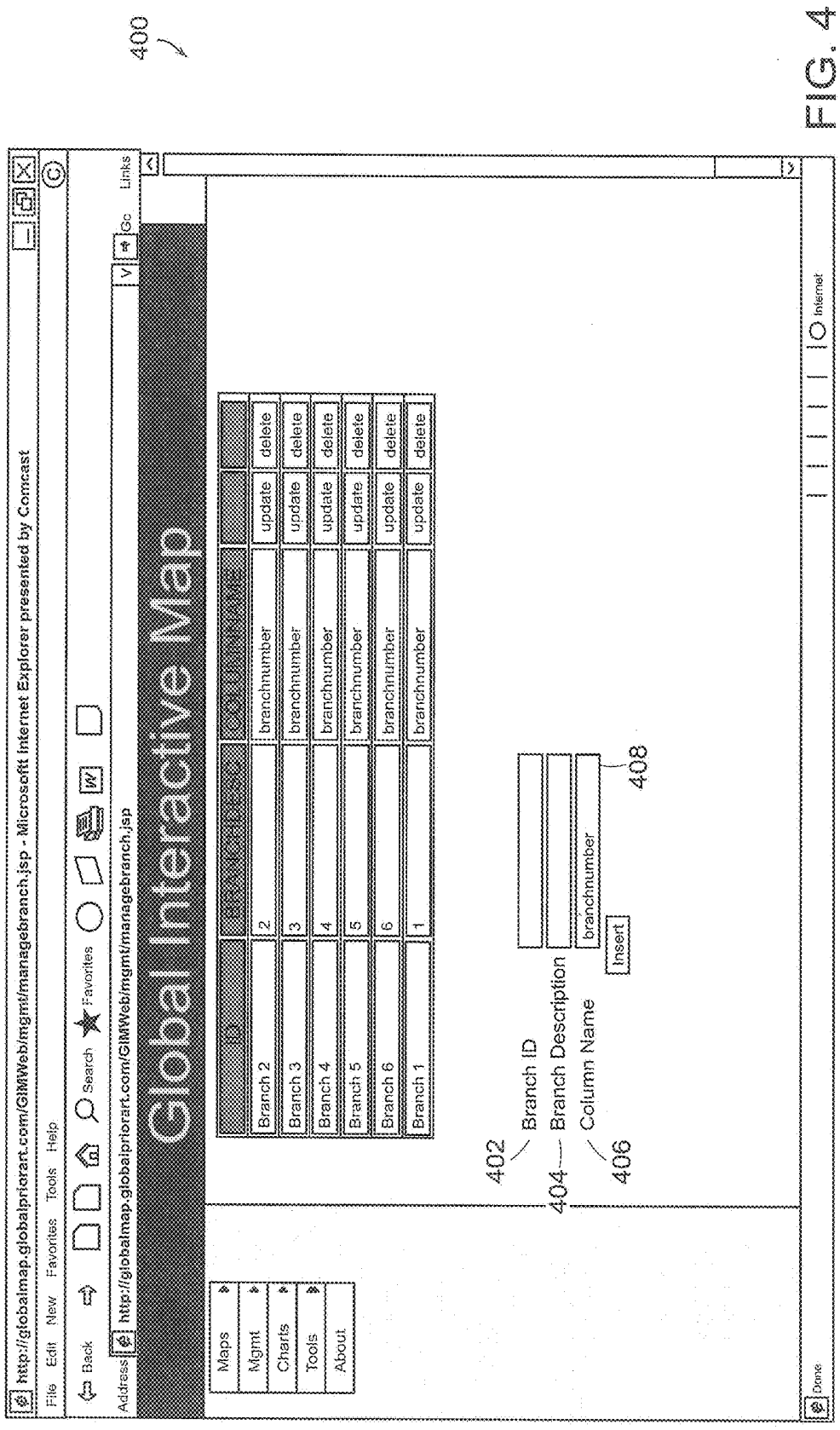
Attribute 1
Attribute 2
Attribute 3
Attribute 4
Attribute 6
Attribute 7
Attribute 8
Attribute 9
Attribute 10

FIG. 3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Branch 3 3 branchnumber update delete
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Branch 5 5 branchnumber update delete
Branch 6 6 branchnumber update delete
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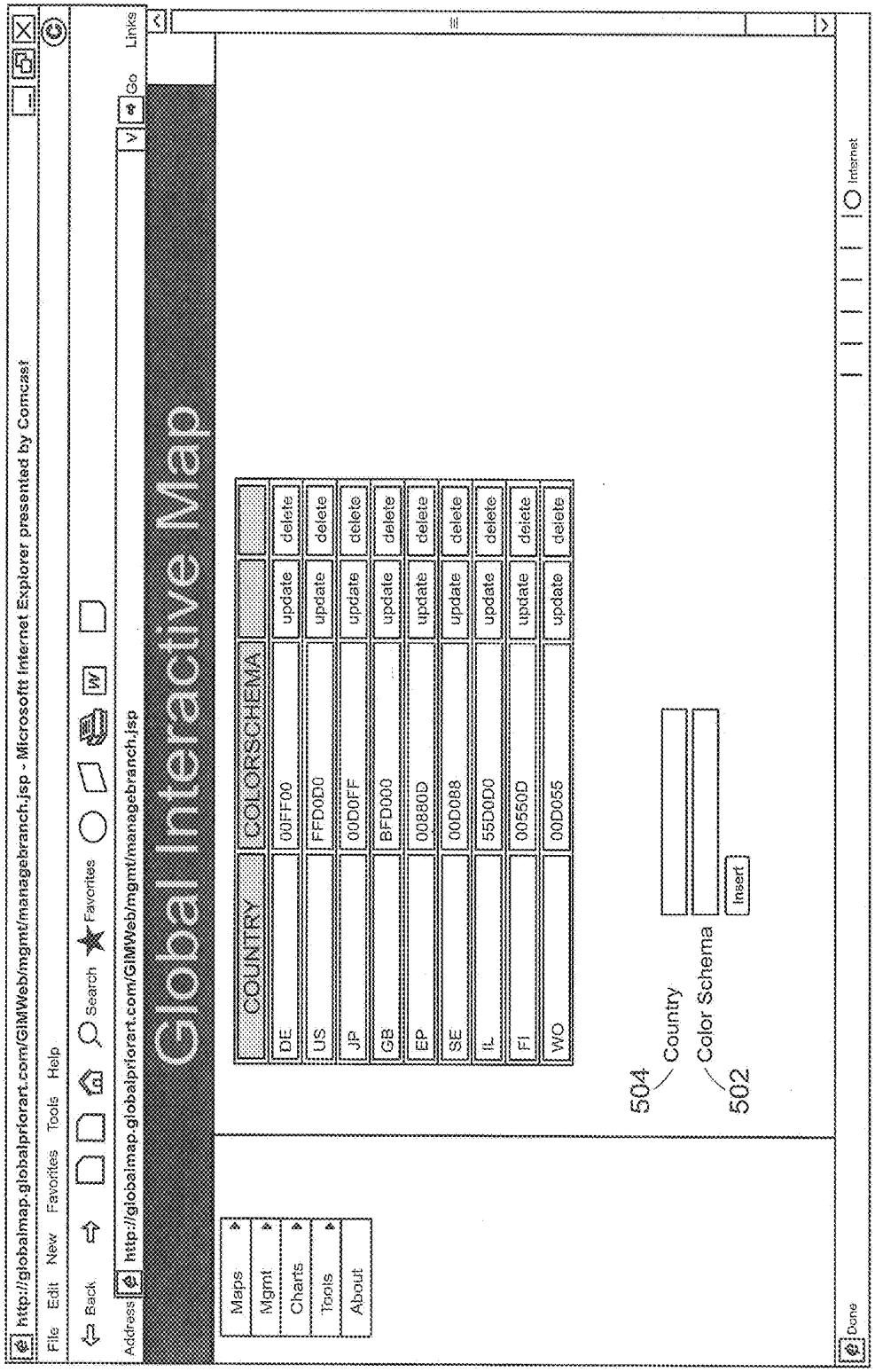
FIG. 4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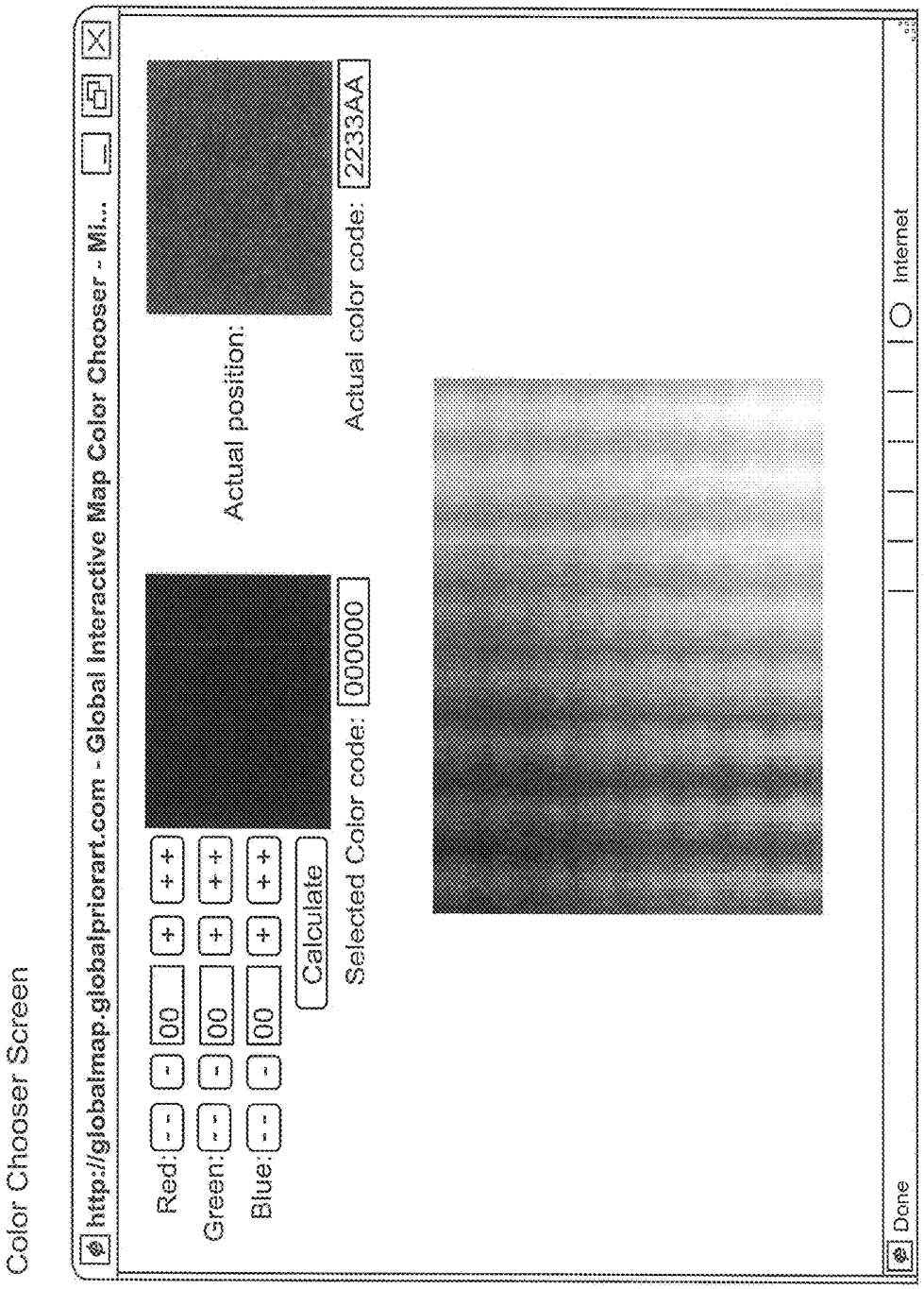
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FIG. 6

700

Sample Data file

"US4809160","US","10/28/1985","10/28/1985","2/28/1989","Privilege level checking instruction for implementing a secure hierarchical computer system","Mahon; Michael J.|Baum; Allen|Bryg; William R.|Miller; Terrence C.","Hewlett-Packard Company","A low overhead way for insuring that only routines of sufficient privilege can execute on a secured page of memory in an hierarchial computer system, and for raising the privilege level of a low privilege process in an orderly and secure way is presented. This is done through the execution of a single ""gateway"" branch instruction standing between a procedure call by a lower privileged routine, such as a user program, and an operating system itself.
"","","2","","","

Sample Layout file

702

PATENTNUMBER
PRIORITYCOUNTRY
PRIORITYDATE
APPLICATIONDATE
PUBLICATIONDATE
TITLE
INVENTORNAME
ASSIGNEEAPPLICANTNAME
ABSTRACT
SUMMARY
BRANCHNUMBER
ATTR_1_Parametric_Response_1central_mgmt
ATTR_2_Parametric_Response_2various_levels
ATTR_3_Parametric_Response_3ordered_levels
ATTR_4_Parametric_Response_4auto_trigger
ATTR_5_Parametric_Response_5manual_trigger
ATTR_6_File_Tracking_1triggered_DirtyOP
ATTR_7_File_Tracking_2record_AC_hashes
ATTR_8_File_Tracking_3central_validatedMI
ATTR_9_File_Tracking_4new_files_xfer_to_center
ATTR_10_File_Tracking_5globally_prop_MI
ATTR_11_File_Tracking_6local_file_track
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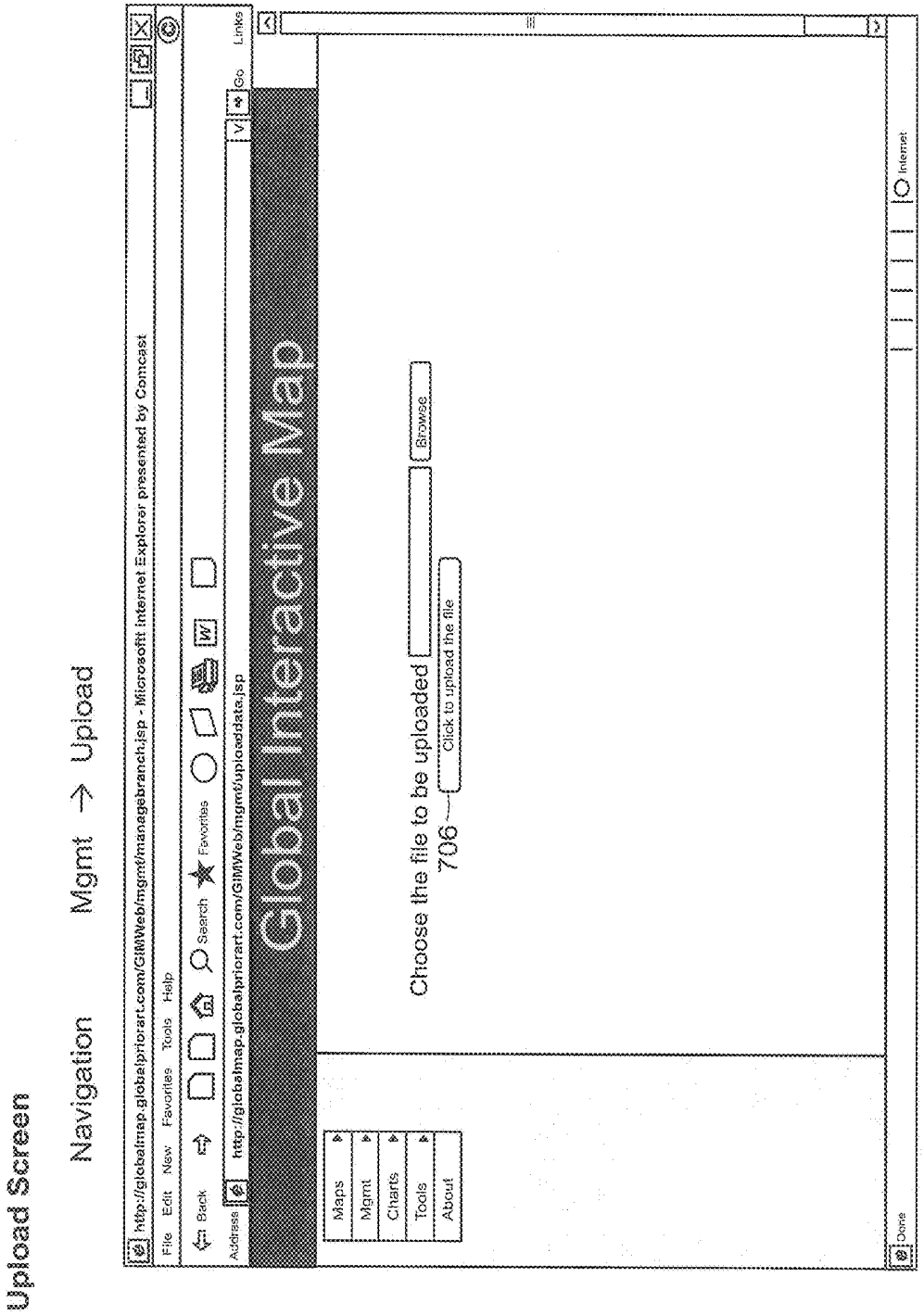
FIG. 7B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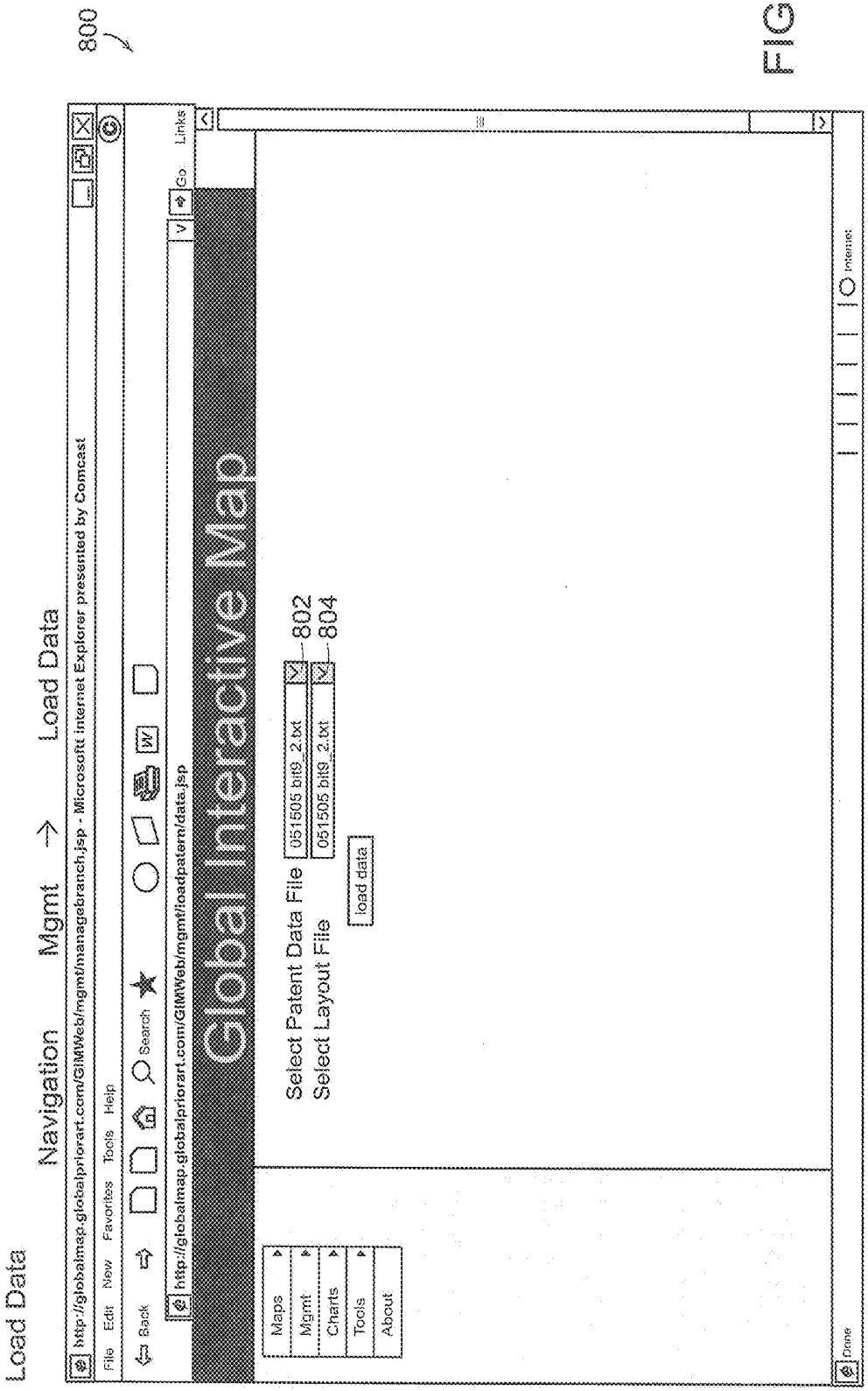
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FIG. 8

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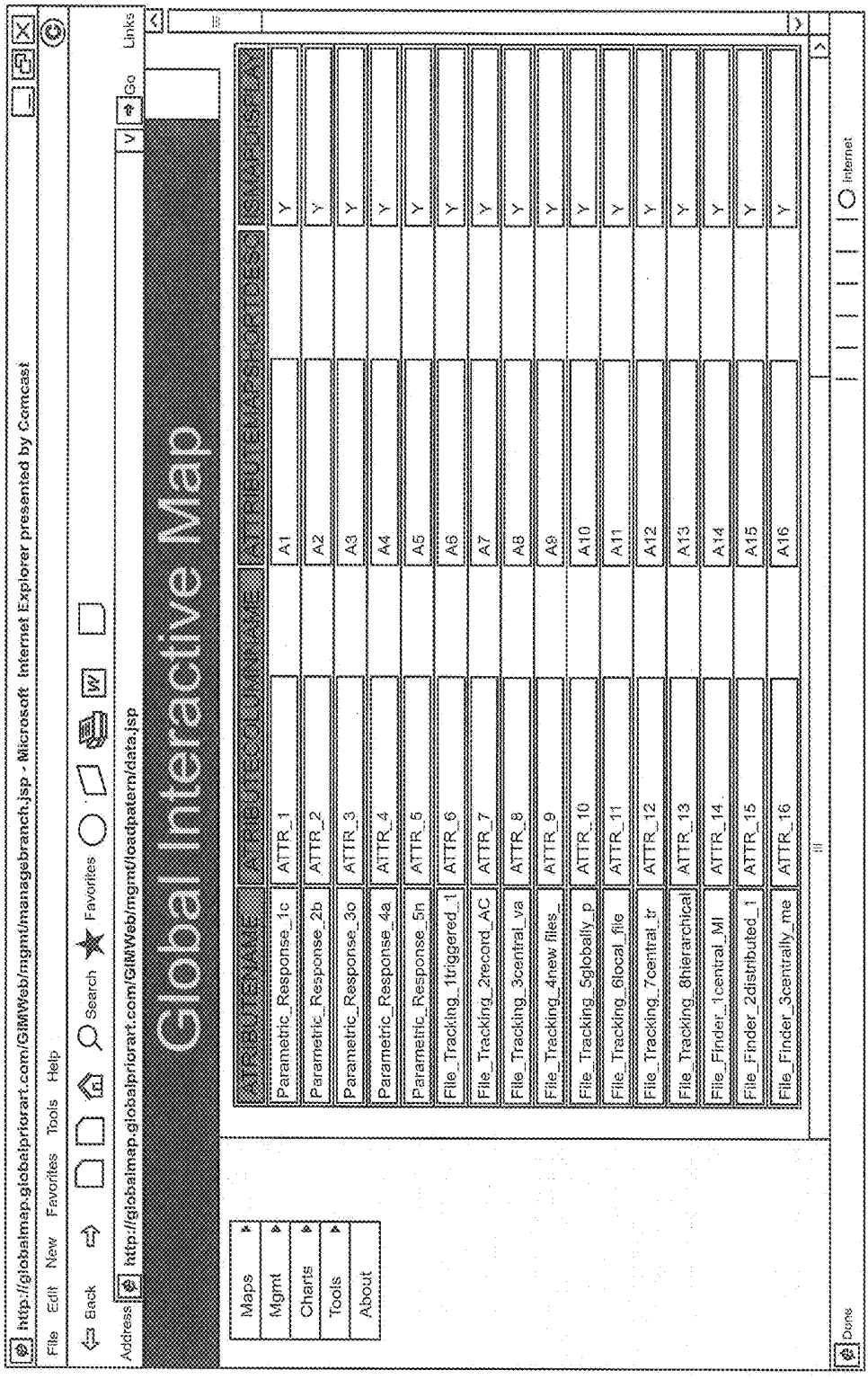
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About
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File_Tracking_4new files_ ATTR_9 A9 Y
File_Tracking_5globally_p ATTR_10 A10 Y
File_Tracking_6local_file ATTR_11 A11 Y
File_Tracking_7central_tr ATTR_12 A12 Y
File_Tracking_8hierarchical ATTR_13 A13 Y
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FIG. 9

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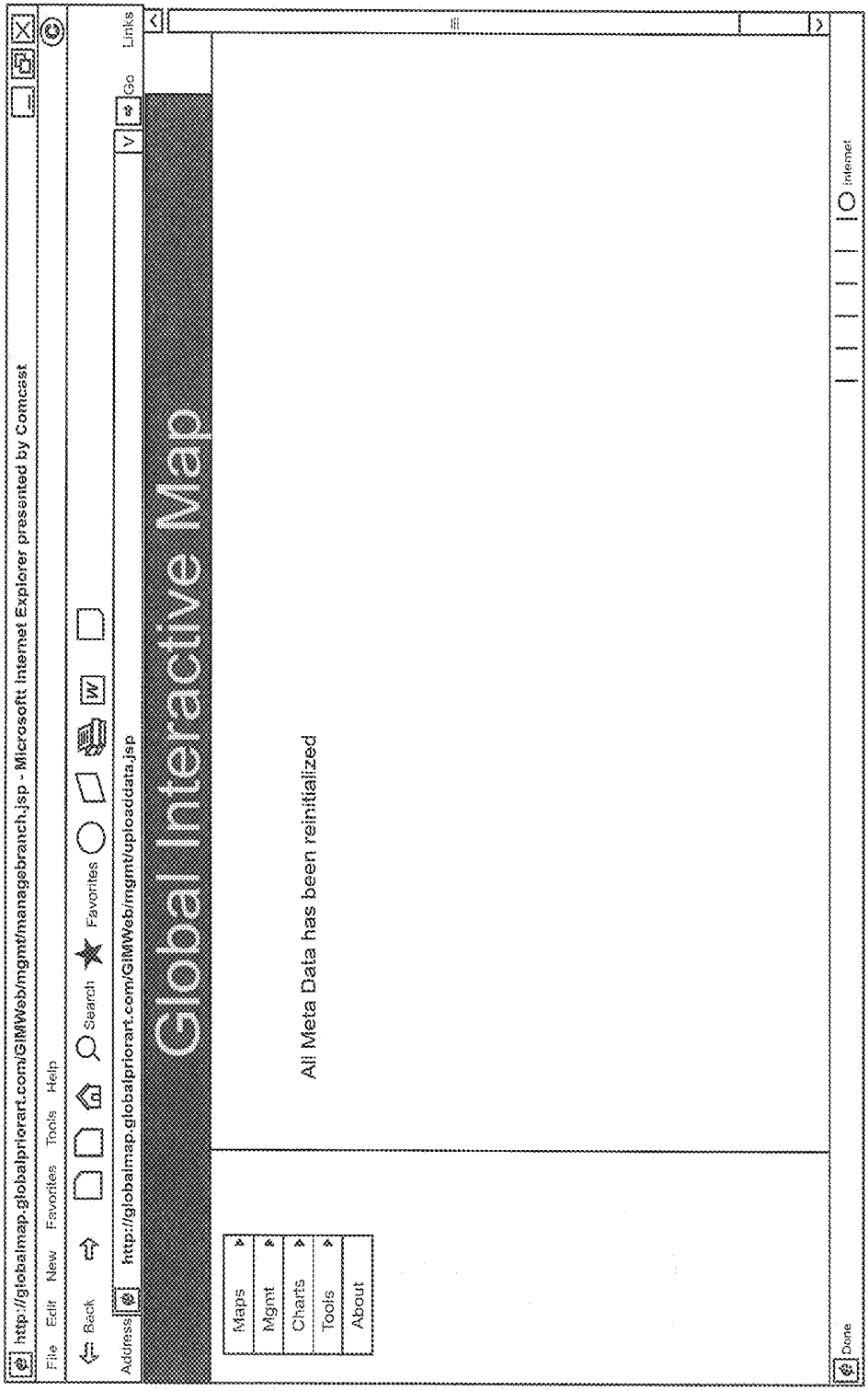
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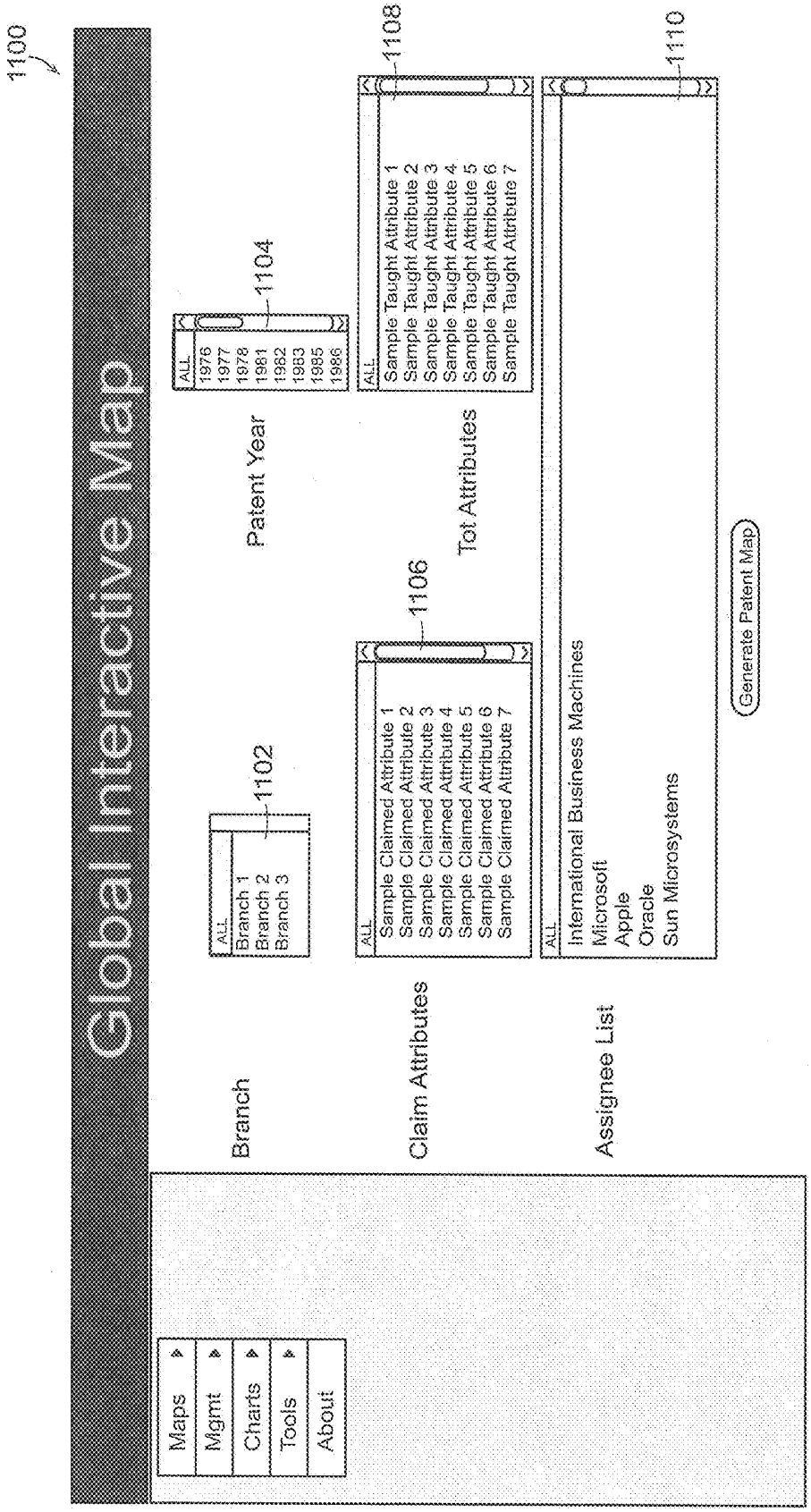
FIG. 10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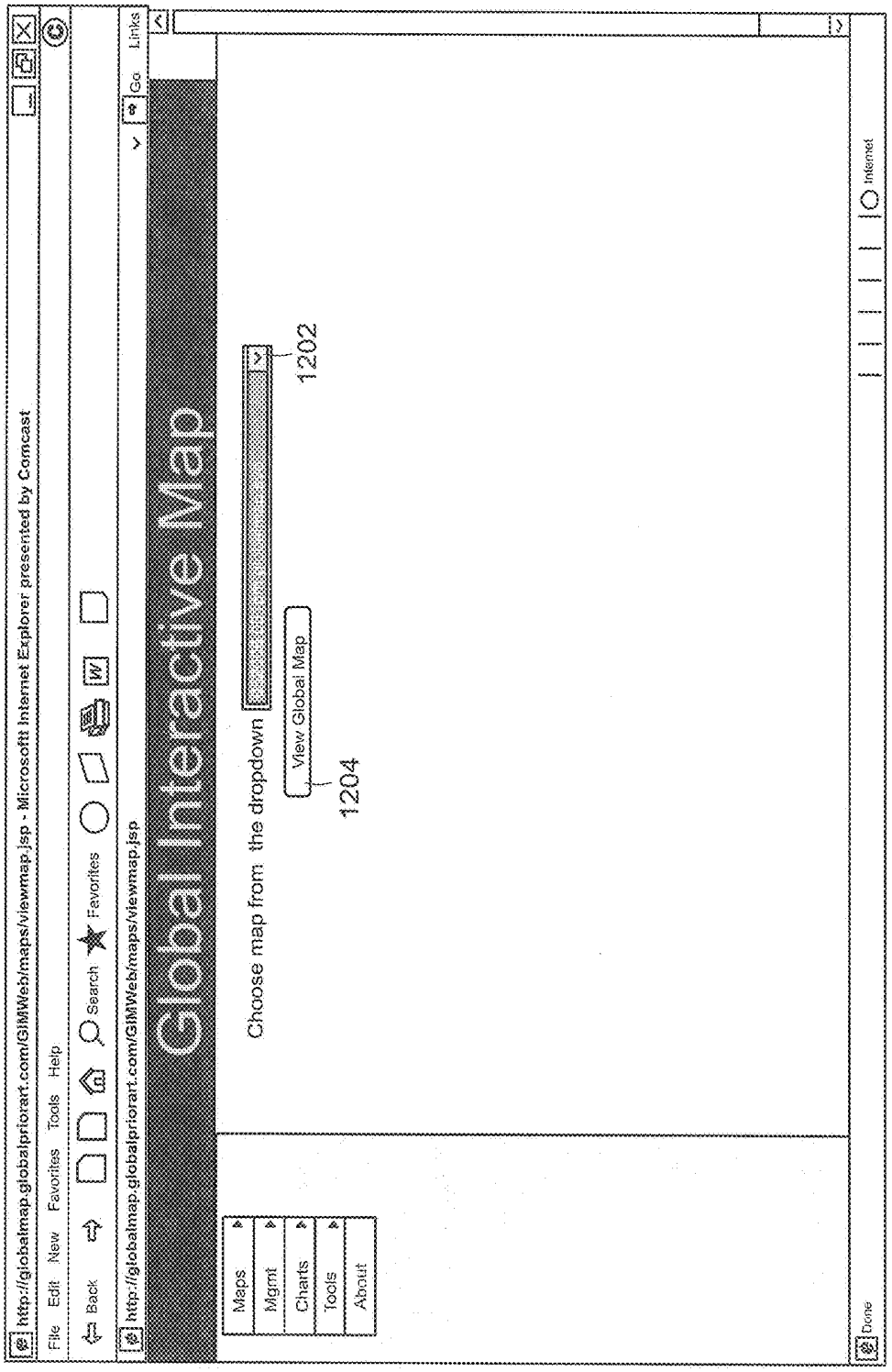
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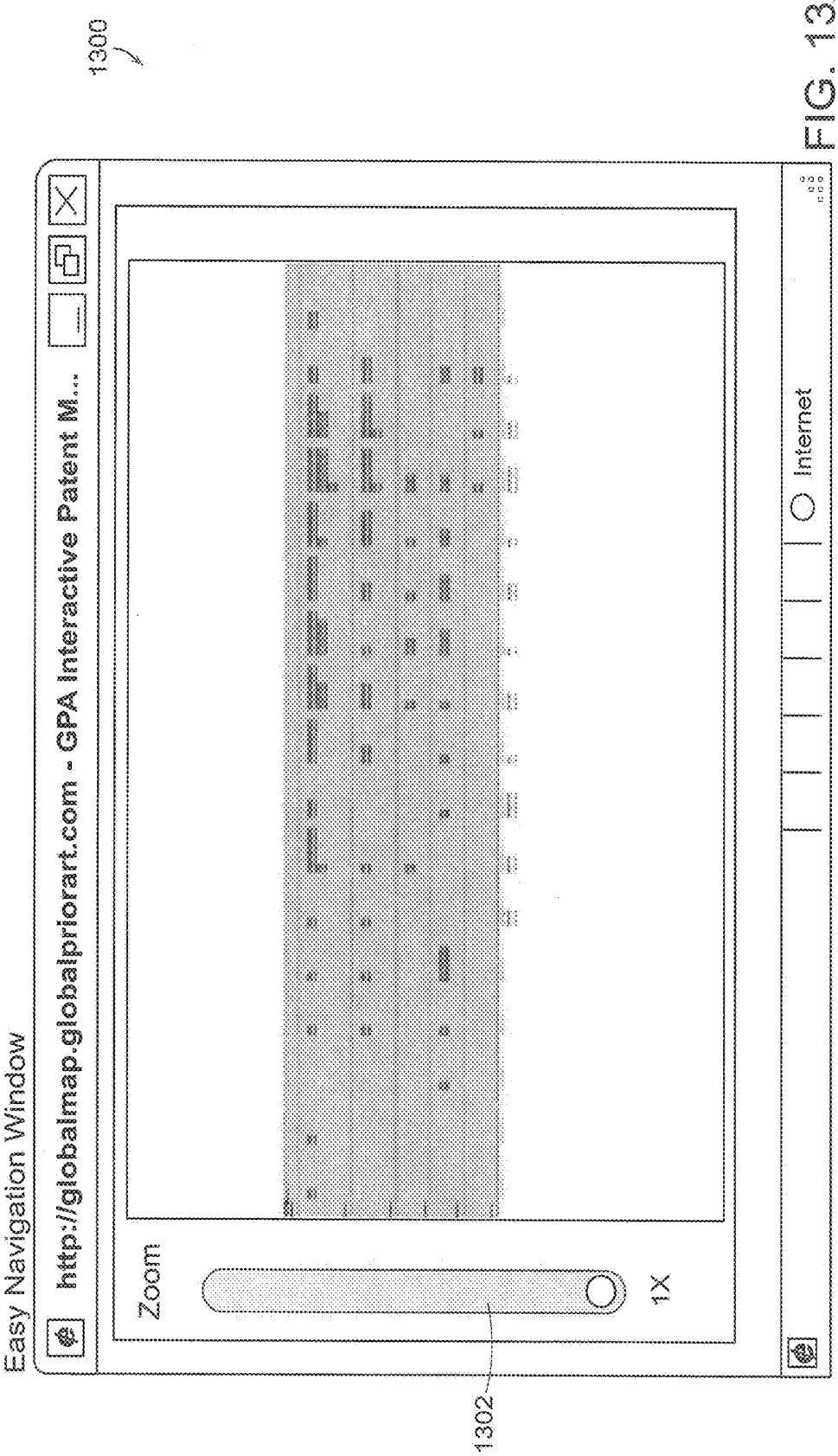
FIG. 12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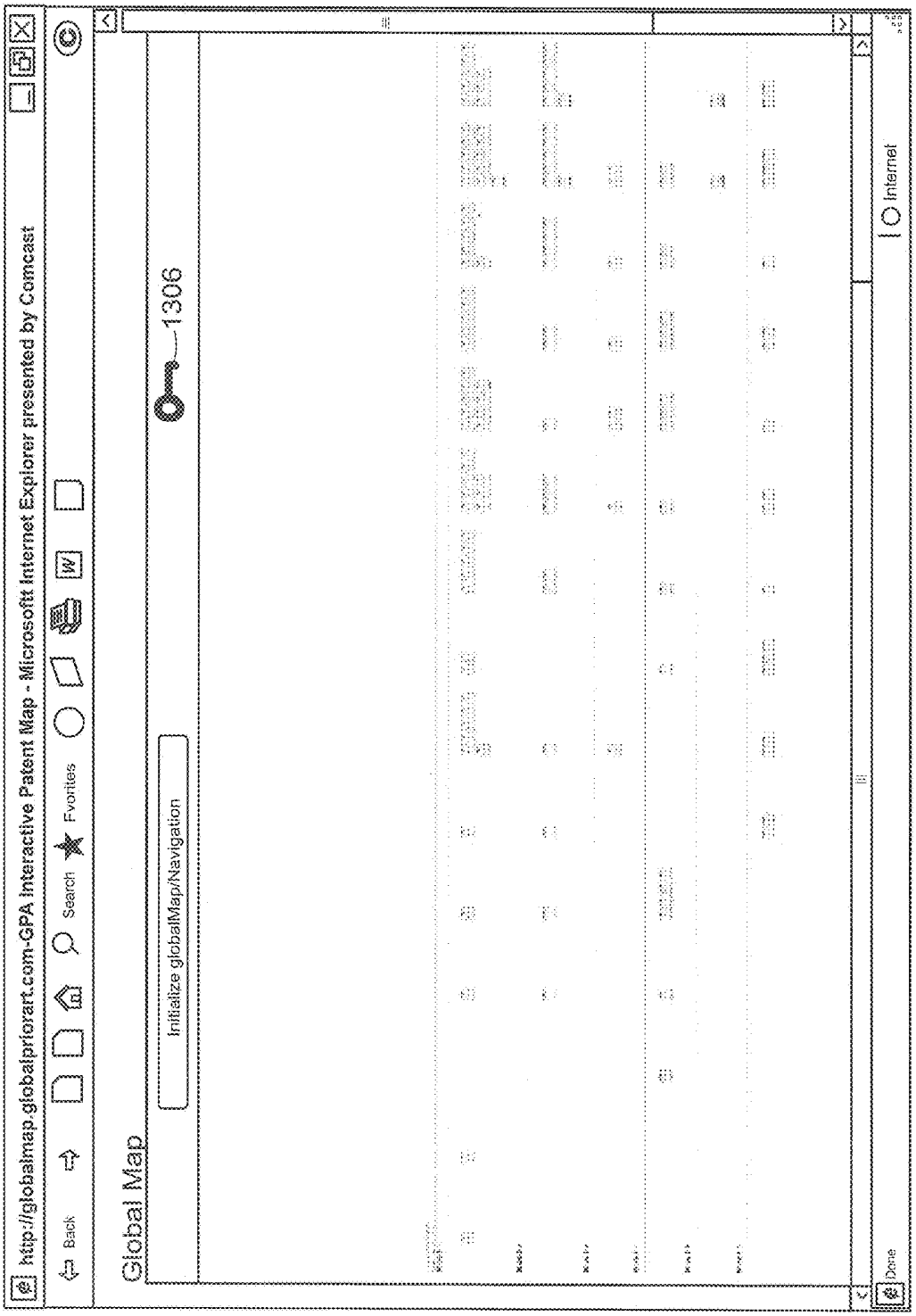
Internet
1302

FIG. 13A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http://globalmap.globalpriorart.com-GPA Interactive Patent Map - Microsoft Internet Explorer presented by Comcast
Back
Search
Favorites
Global Map
Initialize globalMap/Navigation
1306
Done
Internet

FIG. 13B

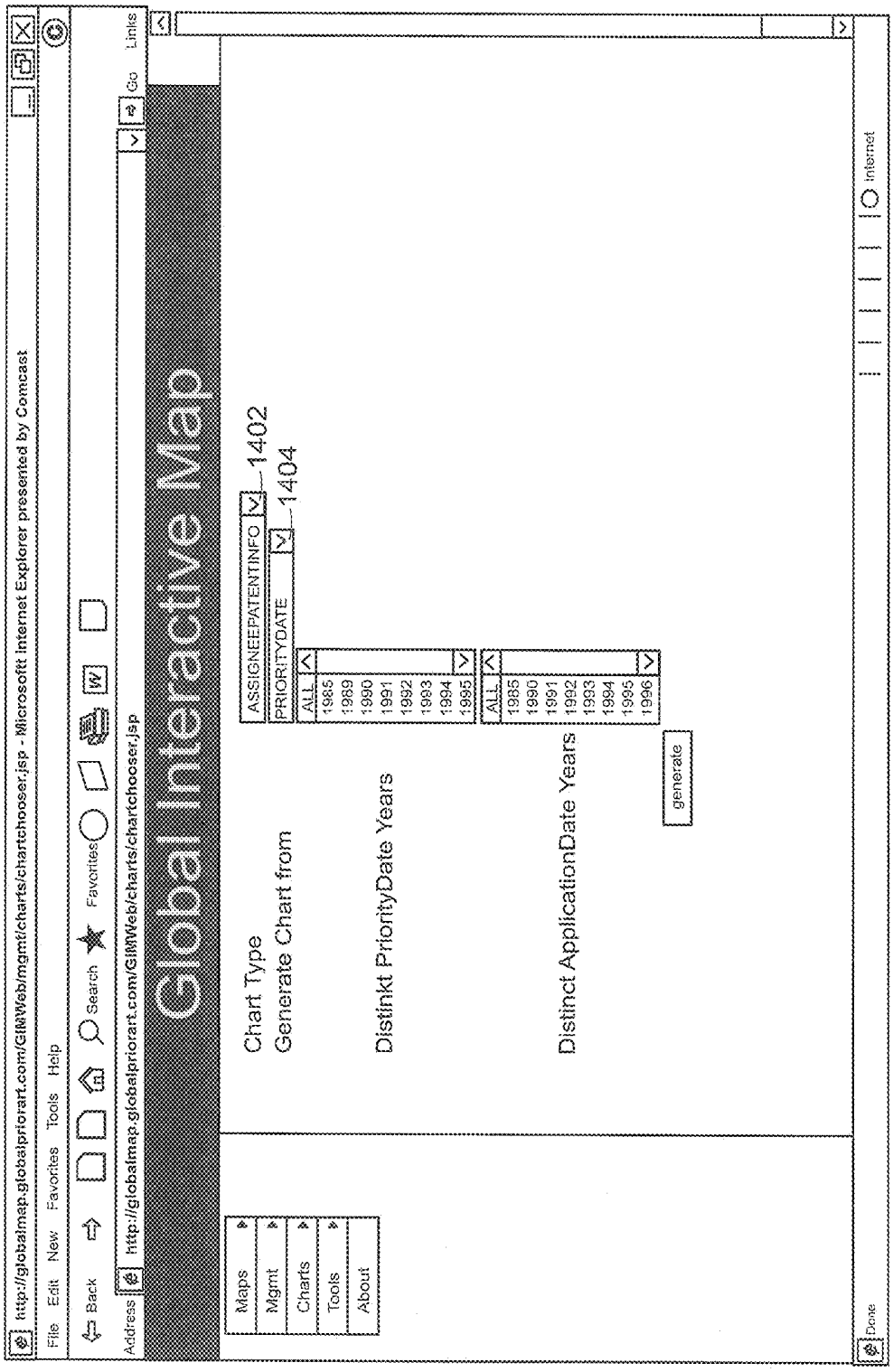
1400
http://globalmap.globalpriorart.com/GIMWeb/mgmt/charts/chartchooser.jsp - Microsoftt Internet Explorer presented by Comcast
File Edit New Favorites Tools Help
Back
Search
Favorites
Address
http://globalmap.globalpriorart.com/GIMWeb/charts/chartchooser.jsp
Go
Links
Global Interactive Map
Maps
Mgmt
Charts
Tools
About
Chart Type
Generate Chart from
ASSIGNEEPATENTINFO
1402
PRIORITYDATE
1404
Distinkt PriorityDate Years
ALL
1985
1989
1990
1991
1992
1993
1994
1995
Distinct ApplicationDate Years
ALL
1985
1990
1991
1992
1993
1994
1995
1996
generate
Done
Internet

FIG. 14

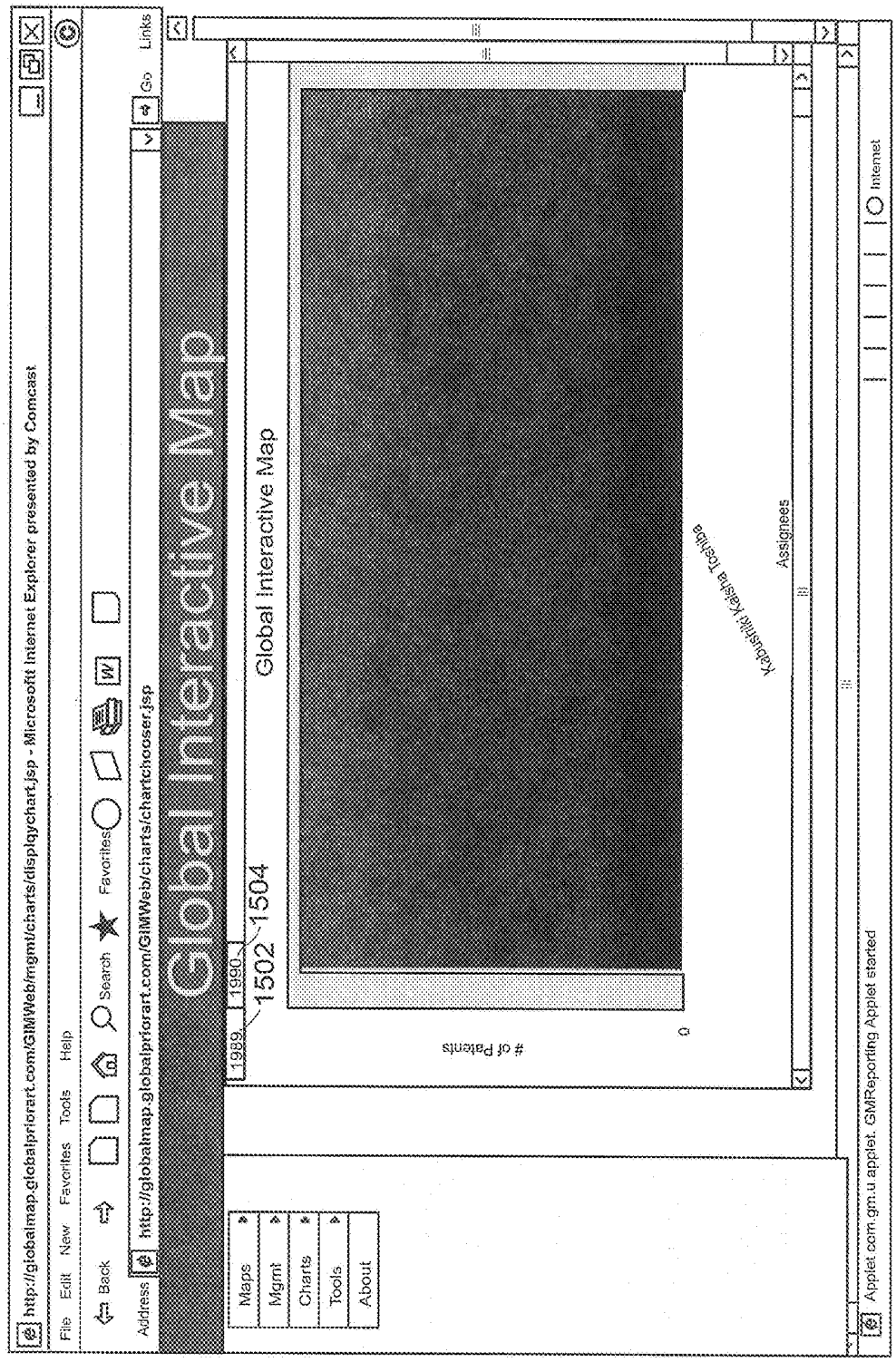
1500
http://globalmap.globalpriorart.com/GIMWeb/mgmt/charts/displaychart.jsp - Microsoft Internet Explorer presented by Comcast
File Edit New Favorites Tools Help
Back
Search
Favorites
Address
http://globalmap.globalpriorart.com/GIMWeb/charts/chartchooser.jsp
Go
Links
Global Interactive Map
Maps
Mgmt
Charts
Tools
About
1989
1990
1502
1504
Global Interactive Map
# of Patents
Kabushiki Kaisha Toshiba
Assignees
Internet
Applet.com.gm.u applet. GMReporting Applet started

FIG. 15

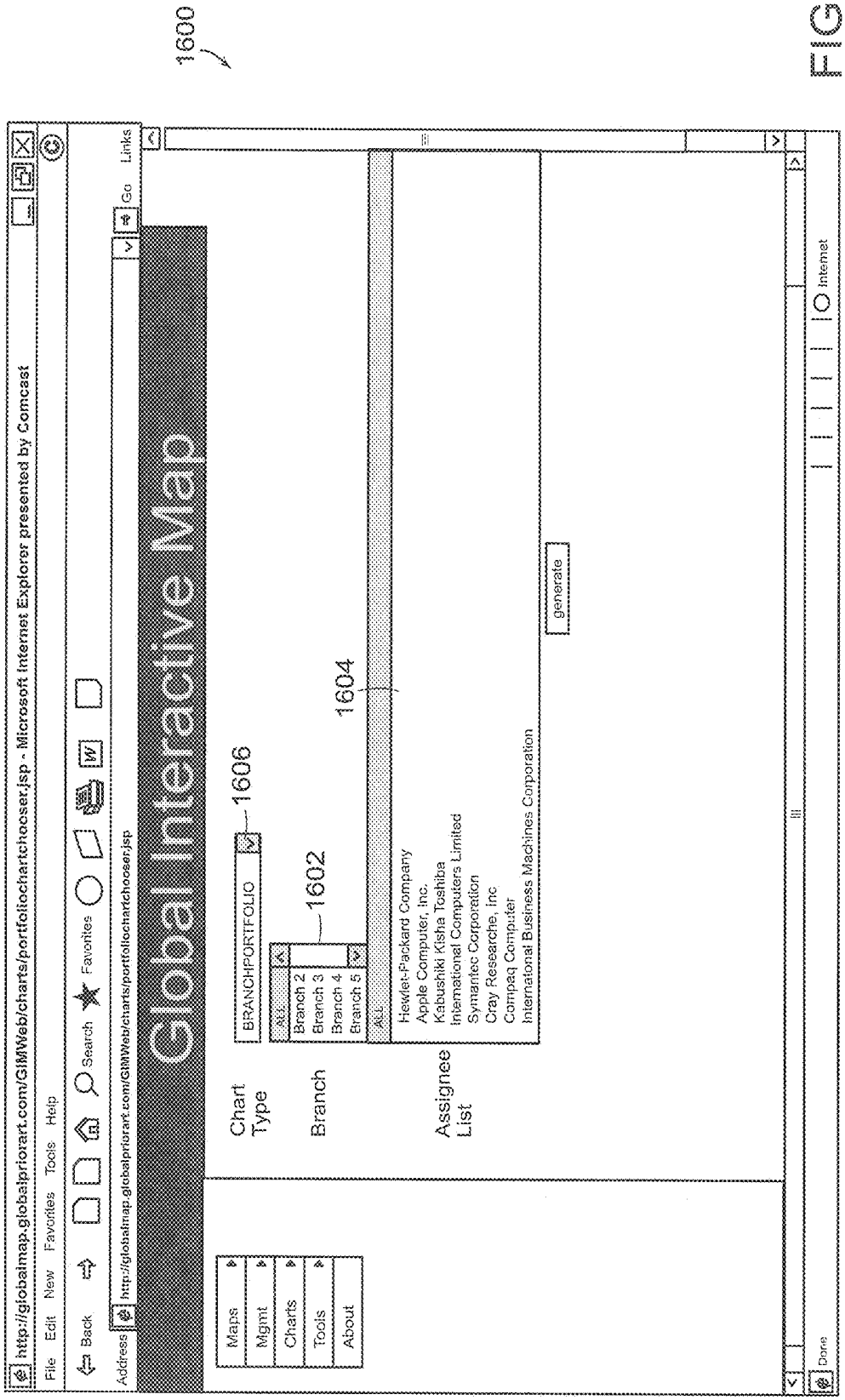
1600
http://globalmap.globalpriorart.com/GIMWeb/charts/portfoliochartchoser.jsp - Microsoft Internet Explorer presented by Comcast
File Edit View Favorites Tools Help
Back Search Favorites
Address http://globalmap.globalpriorart.com/GIMWeb/charts/portfoliochartchoser.jsp
Go Links
Global Interactive Map
Maps
Mgmt
Charts
Tools
About
Chart Type
BRANCHPORTFOLIO
1606
Branch
ALL
Branch 2
Branch 3
Branch 4
Branch 5
1602
1604
Assignee List
ALL
Hewlet-Packard Company
Apple Computer, Inc.
Kabushiki Kisha Toshiba
International Computers Limited
Symantec Corporation
Cray Researche, Inc
Compaq Computer
International Business Machines Corporation
generate
Done
Internet

FIG. 16

1700

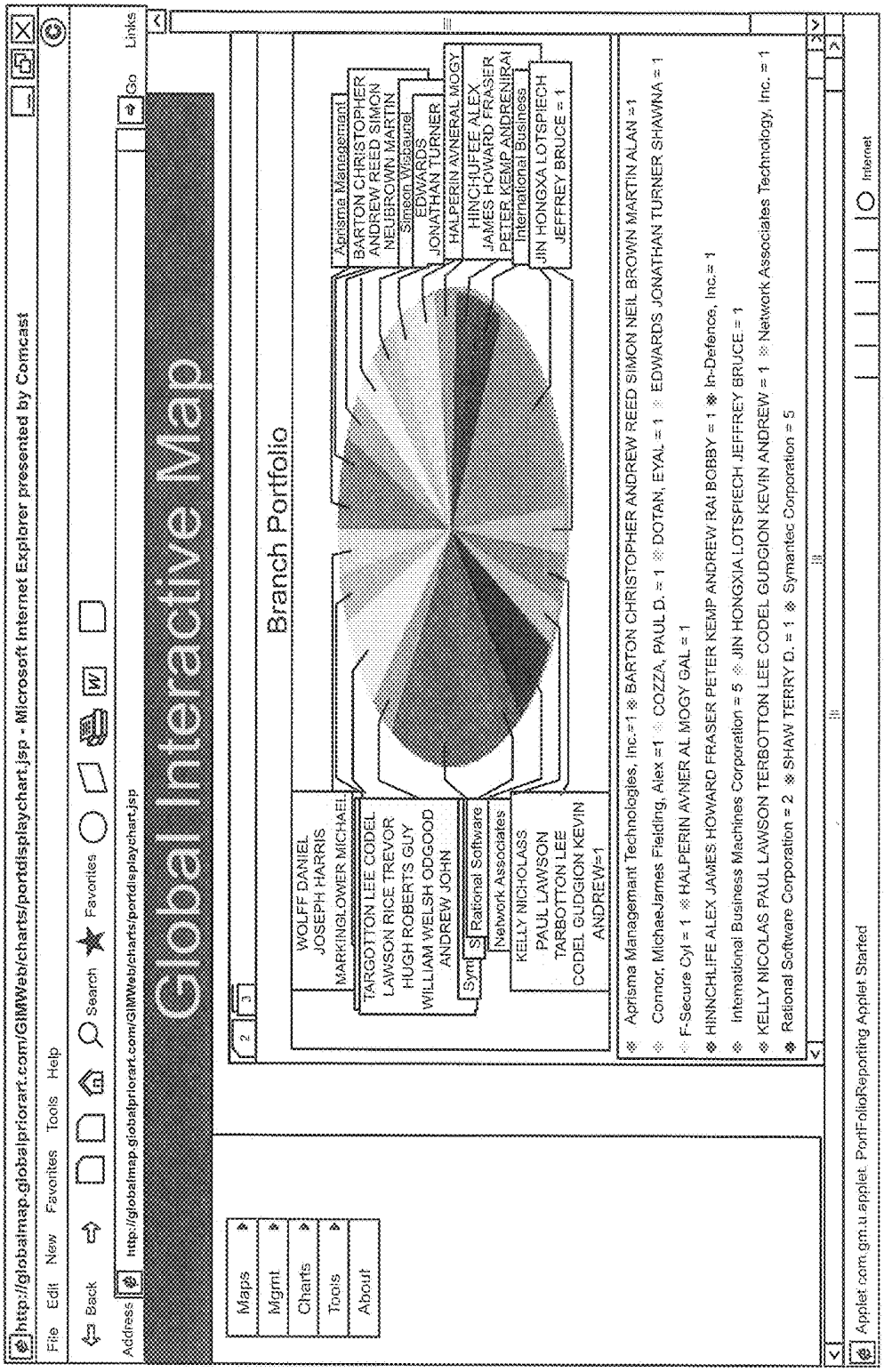
http://globalmap.globalpriorart.com/GIMWeb/charts/portdisplaychart.jsp - Microsoft Internet Explorer presented by Comcast
File Edit New Favorites Tools Help
Back Search Favorites
Address http://globalmap.globalpriorart.com/GIMWeb/charts/portdisplaychart.jsp
Go Links
Global Interactive Map
Maps
Mgmt
Charts
Tools
About
Branch Portfolio
WOLFF DANIEL
JOSEPH HARRIS
MARKINGLOWER MICHAEL
TARGOTTON LEE CODEL
LAWSON RICE TREVOR
HUGH ROBERTS GUY
WILLIAM WELSH ODGOOD
ANDREW JOHN
Symi
Rational Software
Network Associates
KELLY NICHOLASS
PAUL LAWSON
TARBOTTON LEE
CODEL GUDGION KEVIN
ANDREW=1
Aprisma Management
BARTON CHRISTOPHER
ANDREW REED SIMON
NEUBROWN MARTIN
Simeon Wisbaunel
EDWARDS
JONATHAN TURNER
HALPERIN AVNERAL MOGY
HINCHUFEE ALEX
JAMES HOWARD FRASER
PETER KEMP ANDRENIRAI
International Business
JIN HONGXA LOTSPIECH
JEFFREY BRUCE = 1
Aprisma Management Technologies, Inc.=1 BARTON CHRISTOPHER ANDREW REED SIMON NEIL BROWN MARTIN ALAN =1
Connor, MichaeJames Fielding, Alex =1 COZZA, PAUL D. = 1 DOTAN, EYAL = 1 EDWARDS JONATHAN TURNER SHAWNA = 1
F-Secure Cyl = 1 HALPERIN AVNER AL MOGY GAL = 1
HINNCHLIFE ALEX JAMES HOWARD FRASER PETER KEMP ANDREW RAI BGBBY = 1 In-Defence, Inc.= 1
International Business Machines Corporation = 5 JIN HONGXIA LOTSPIECH JEFFREY BRUCE = 1
KELLY NICOLAS PAUL LAWSON TERBOTTON LEE CODEL GUDGION KEVIN ANDREW = 1 Network Associates Technology, Inc. = 1
Rational Software Corporation = 2 SHAW TERRY D. = 1 Symantec Corporation = 5
Applet com.gm.u.applet. PortFolioReporting Applet Started
Internet

FIG. 17

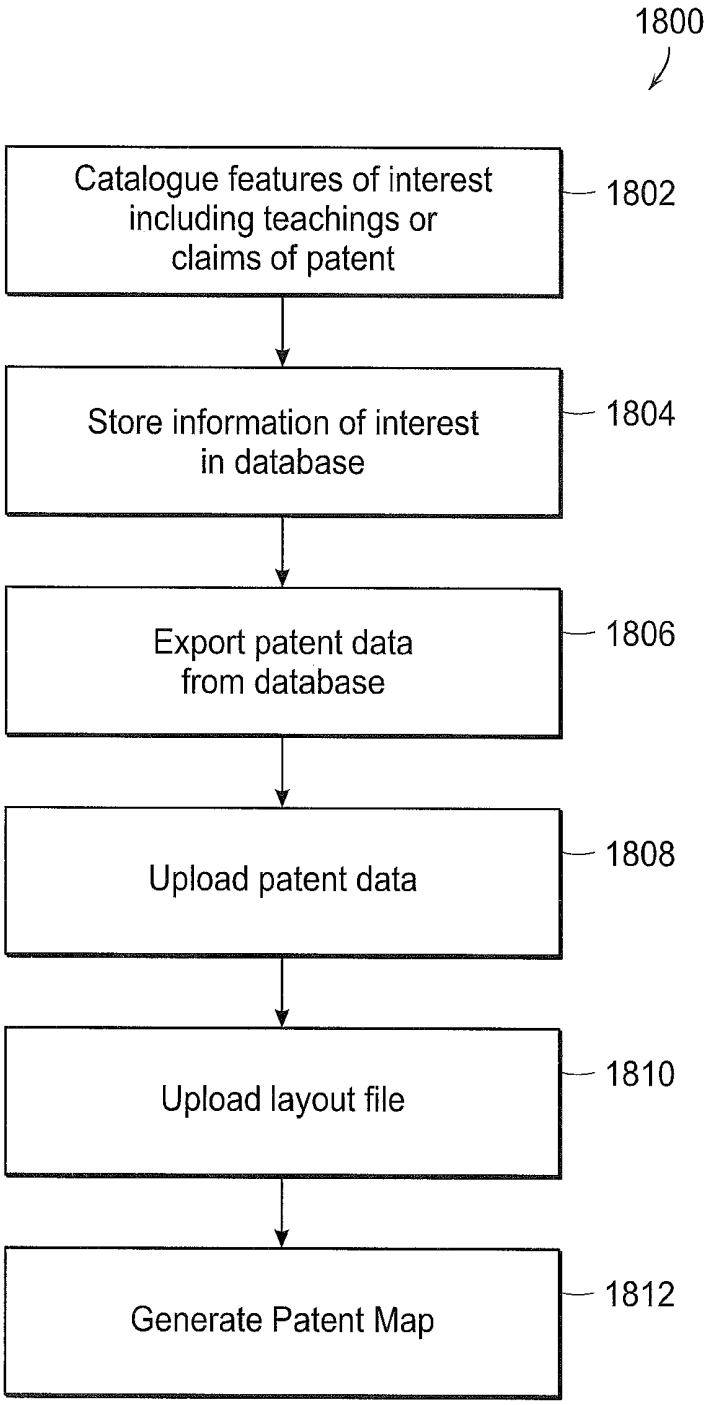
1800
Catalogue features of interest including teachings or claims of patent
1802
Store information of interest in database
1804
Export patent data from database
1806
Upload patent data
1808
Upload layout file
1810
Generate Patent Map
1812

FIG. 18

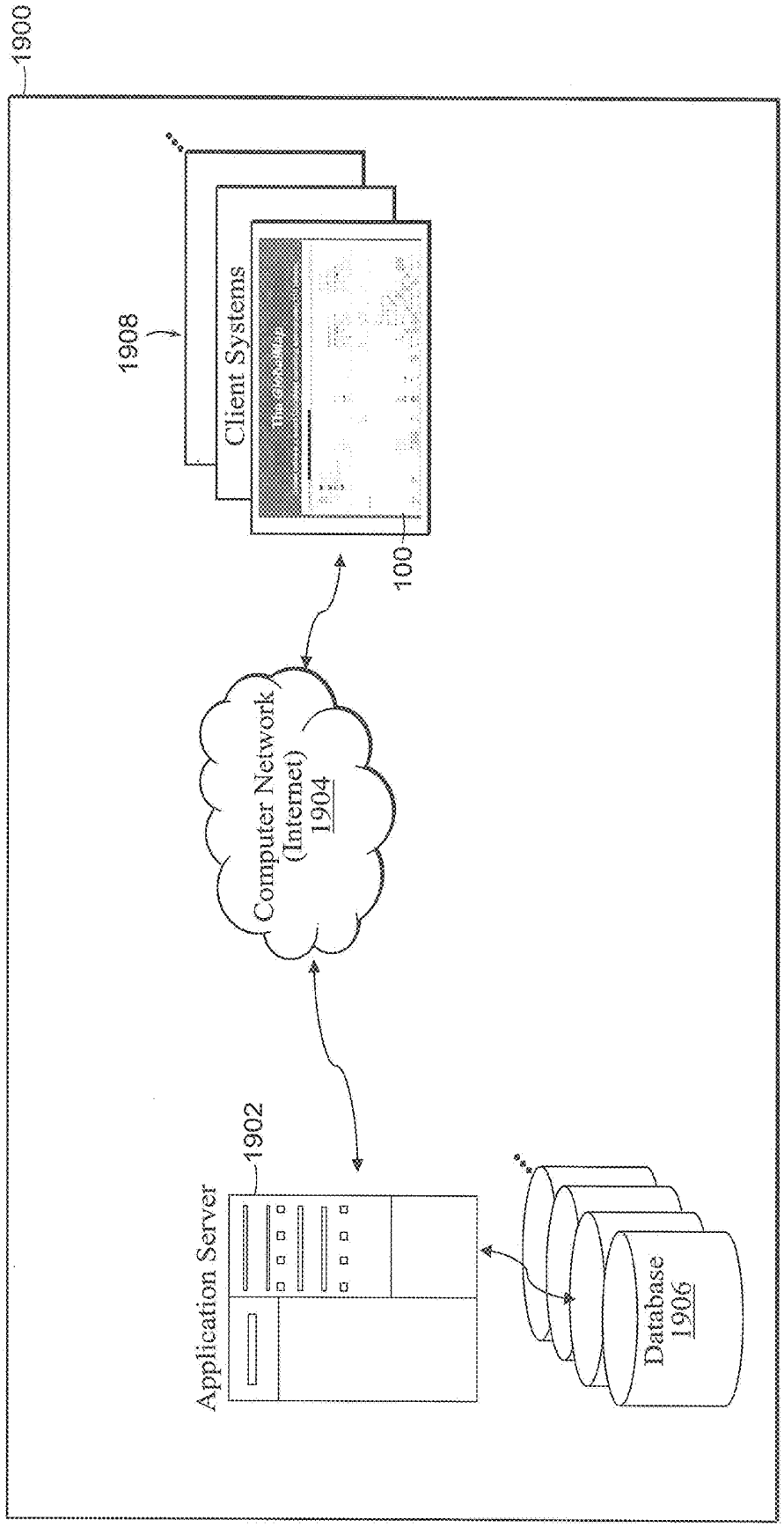
1900
1908
Client Systems
100
Computer Network
(Internet)
1904
1902
Application Server
Database
1906

FIG. 19

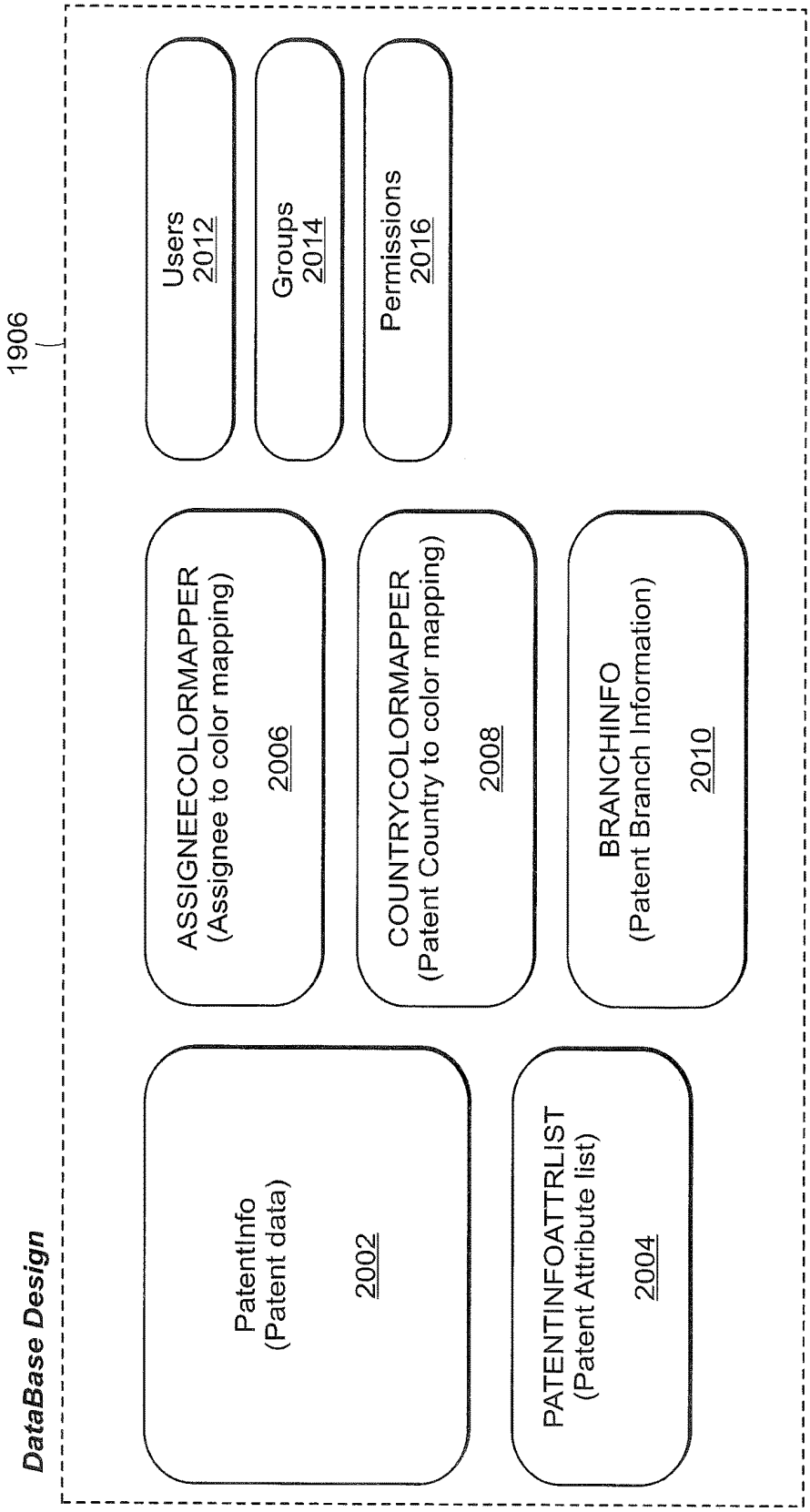
DataBase Design
1906
PatentInfo
(Patent data)
2002
ASSIGNEECOLORMAPPER
(Assignee to color mapping)
2006
Users
2012
PATENTINFOATTRLIST
(Patent Attribute list)
2004
COUNTRYCOLORMAPPER
(Patent Country to color mapping)
2008
Groups
2014
BRANCHINFO
(Patent Branch Information)
2010
Permissions
2016

FIG. 20

Map Generation and Static Data Maintenance

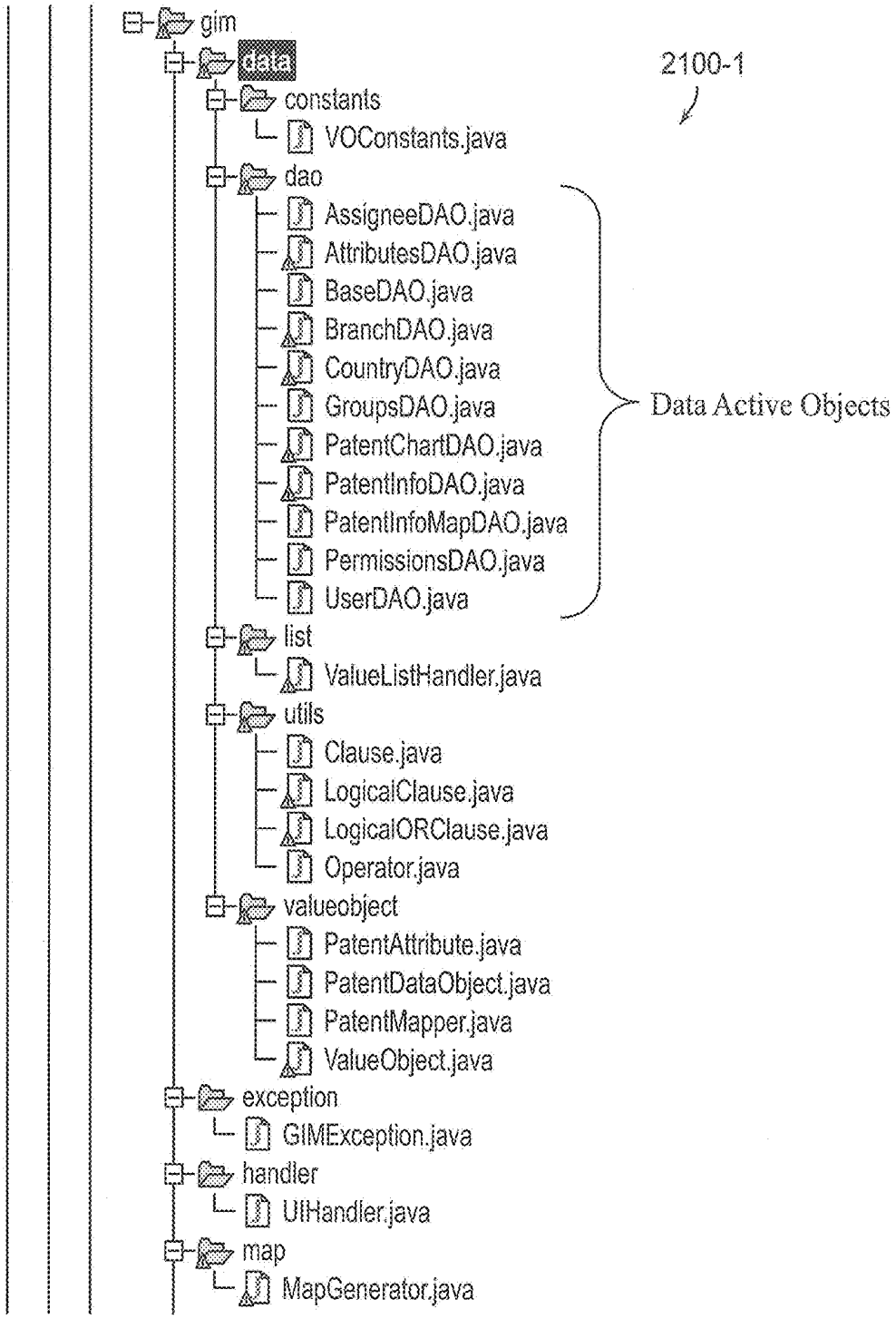

FIG. 21A

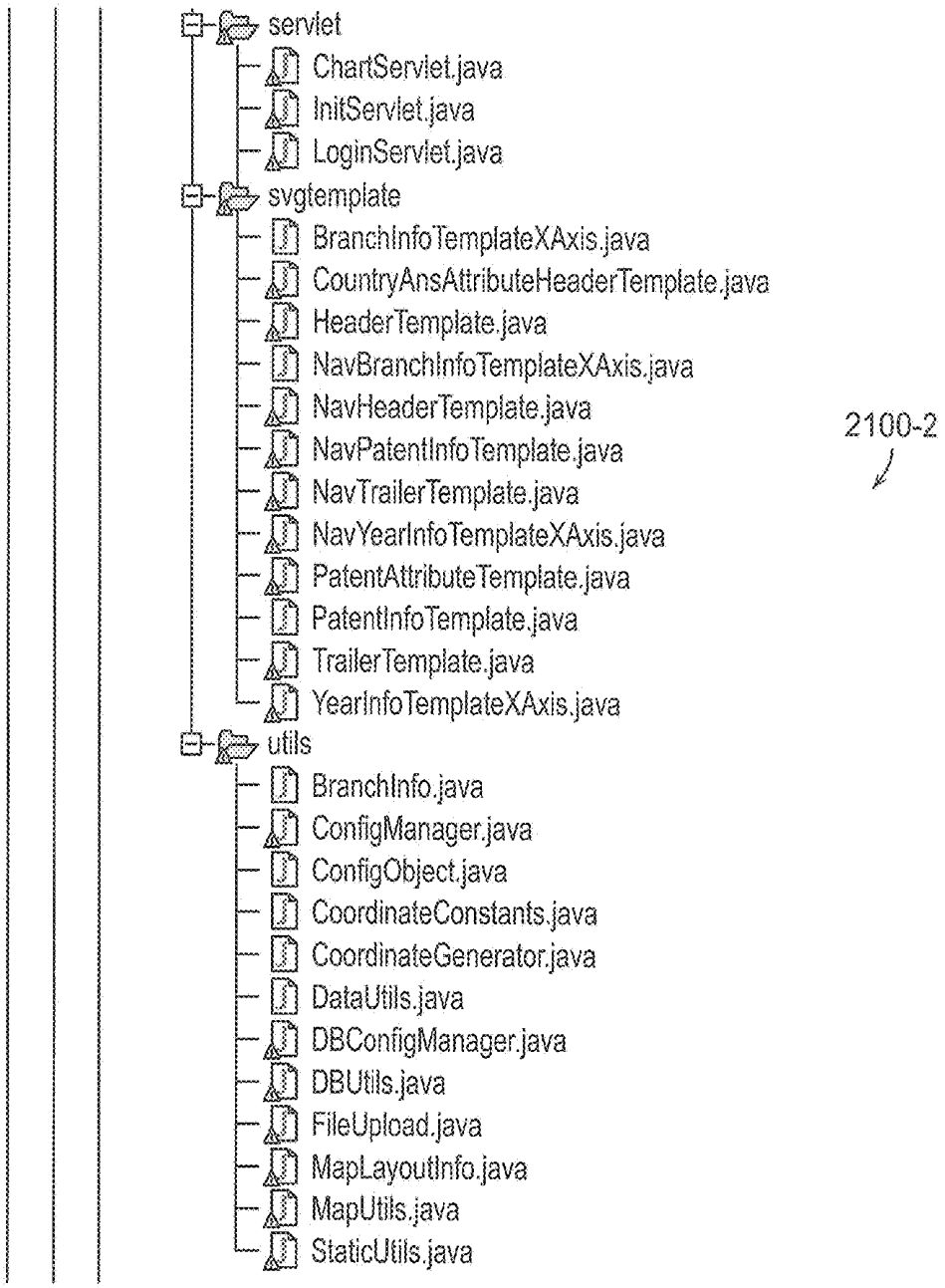
servlet
ChartServlet.java
InitServlet.java
LoginServlet.java
svgtemplate
BranchInfoTemplateXAxis.java
CountryAnsAttributeHeaderTemplate.java
HeaderTemplate.java
NavBranchInfoTemplateXAxis.java
NavHeaderTemplate.java
NavPatentInfoTemplate.java
NavTrailerTemplate.java
NavYearInfoTemplateXAxis.java
PatentAttributeTemplate.java
PatentInfoTemplate.java
TrailerTemplate.java
YearInfoTemplateXAxis.java
utils
BranchInfo.java
ConfigManager.java
ConfigObject.java
CoordinateConstants.java
CoordinateGenerator.java
DataUtils.java
DBConfigManager.java
DBUtils.java
FileUpload.java
MapLayoutInfo.java
MapUtils.java
StaticUtils.java
2100-2

FIG. 21B

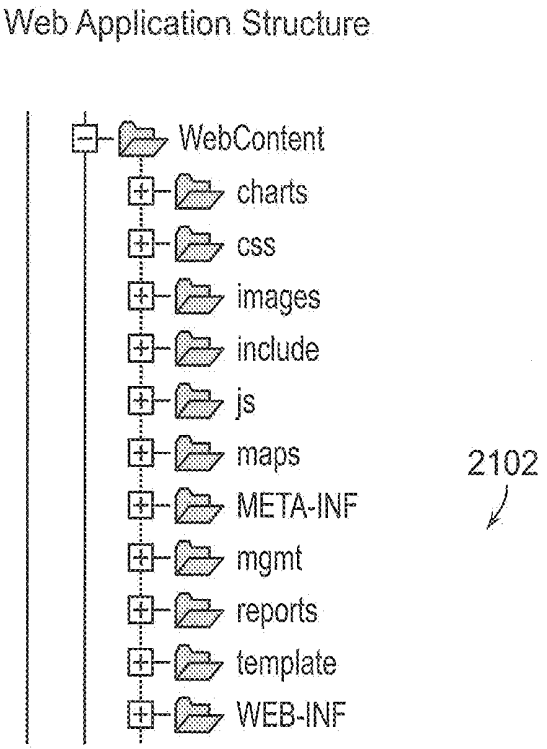
Web Application Structure
WebContent
charts
css
images
include
js
maps
META-INF
mgmt
reports
template
WEB-INF
2102

FIG. 21C

Map Templates

- attributeidtemplate.vim
- branchinfooyaxis.vim
- countryidtemplate.vim
- gblmap.properties
- gia-config.xml
- gia-rules.xml
- headertemplate.vim
- navbranchinfoyaxis.vim
- navheadertemplate.vim
- navpatentinfo.vim
- navtrailertemplate.vim
- navyearinfoxaxis.vim
- patentinfo.vim
- trailertemplate.vim
- velocity.properties
- yearinfoxaxis.vim

2104

FIG. 21D

UI Chart Structure

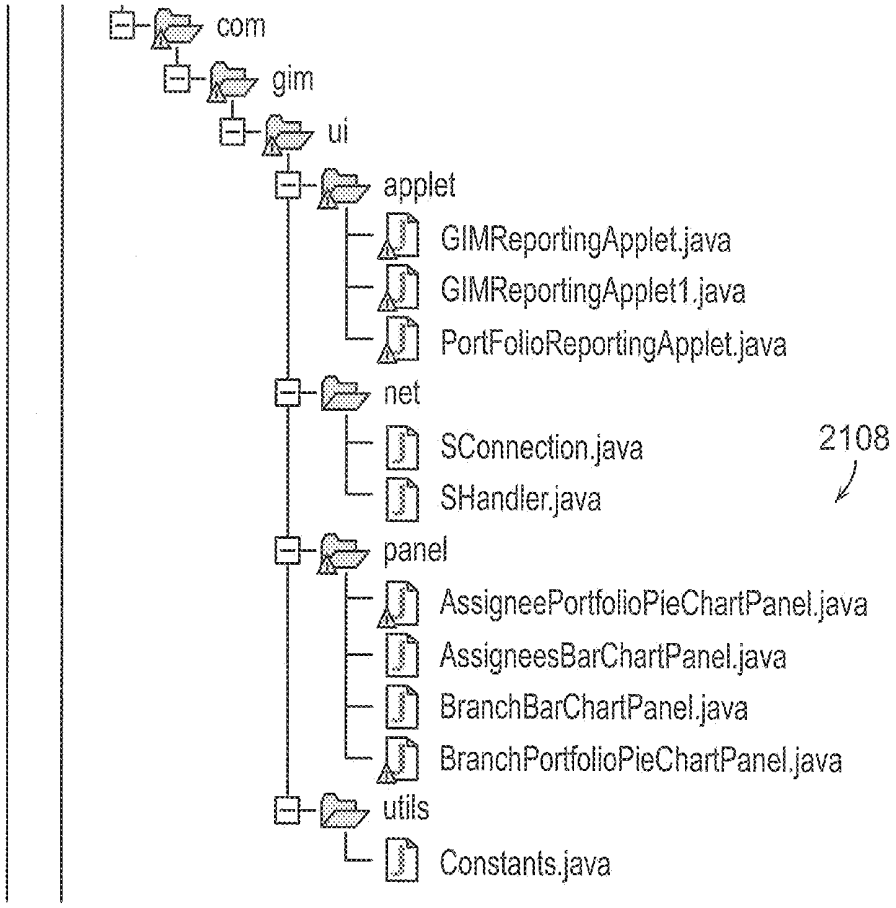

FIG. 21E

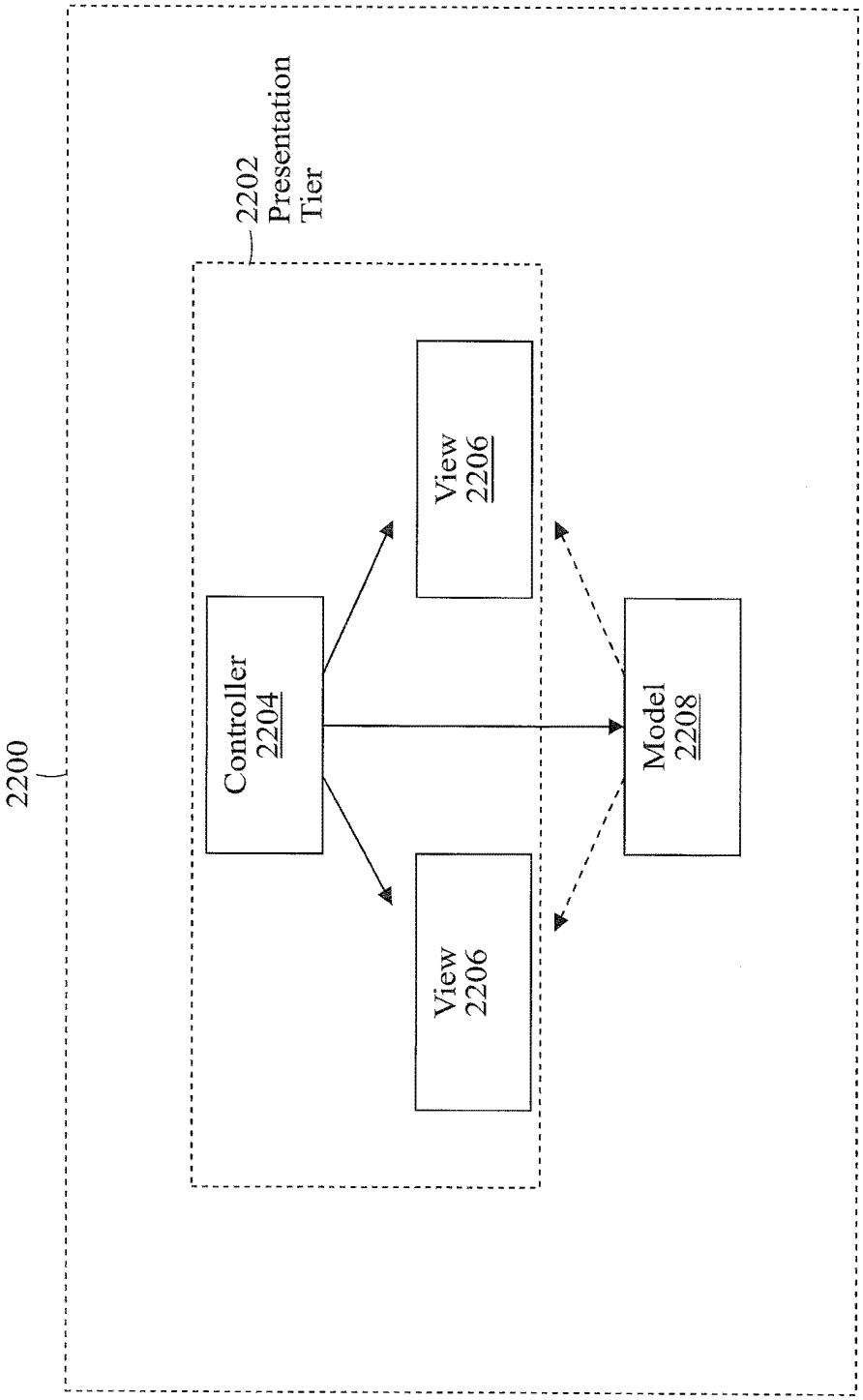
2200
2202
Presentation
Tier
Controller
2204
View
2206
View
2206
Model
2208

FIG. 22

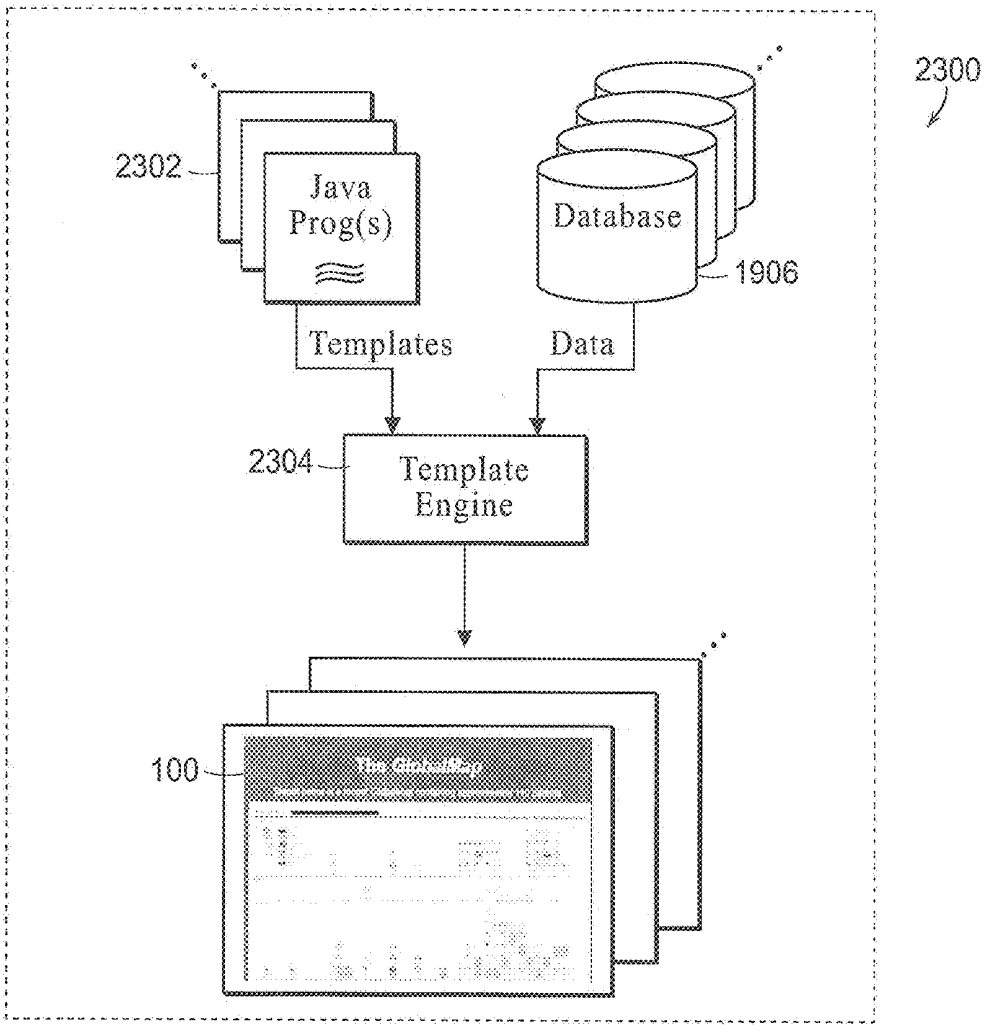
2300
2302
Java
Prog(s)
Database
1906
Templates
Data
2304
Template
Engine
100

FIG. 23

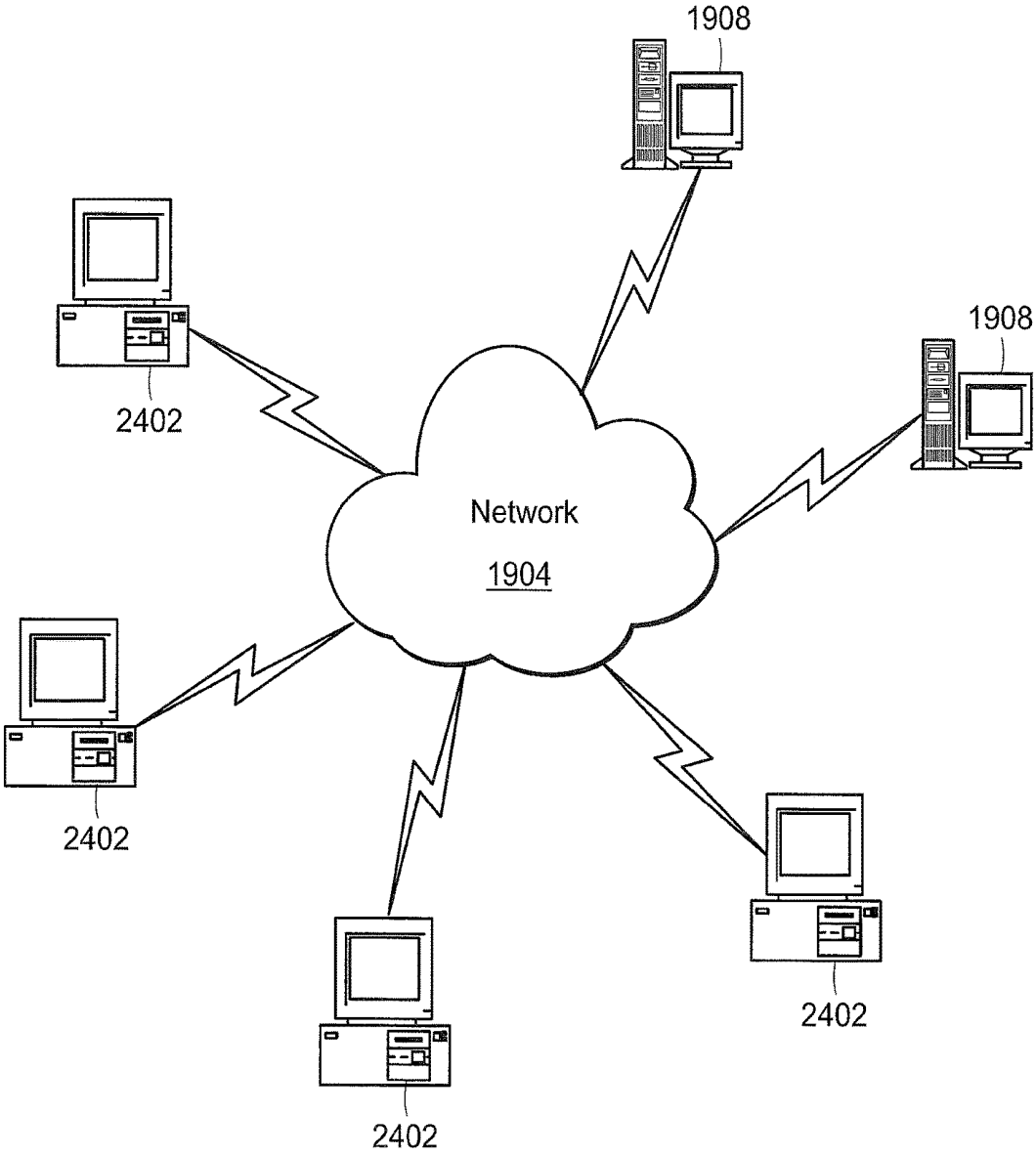

FIG. 24

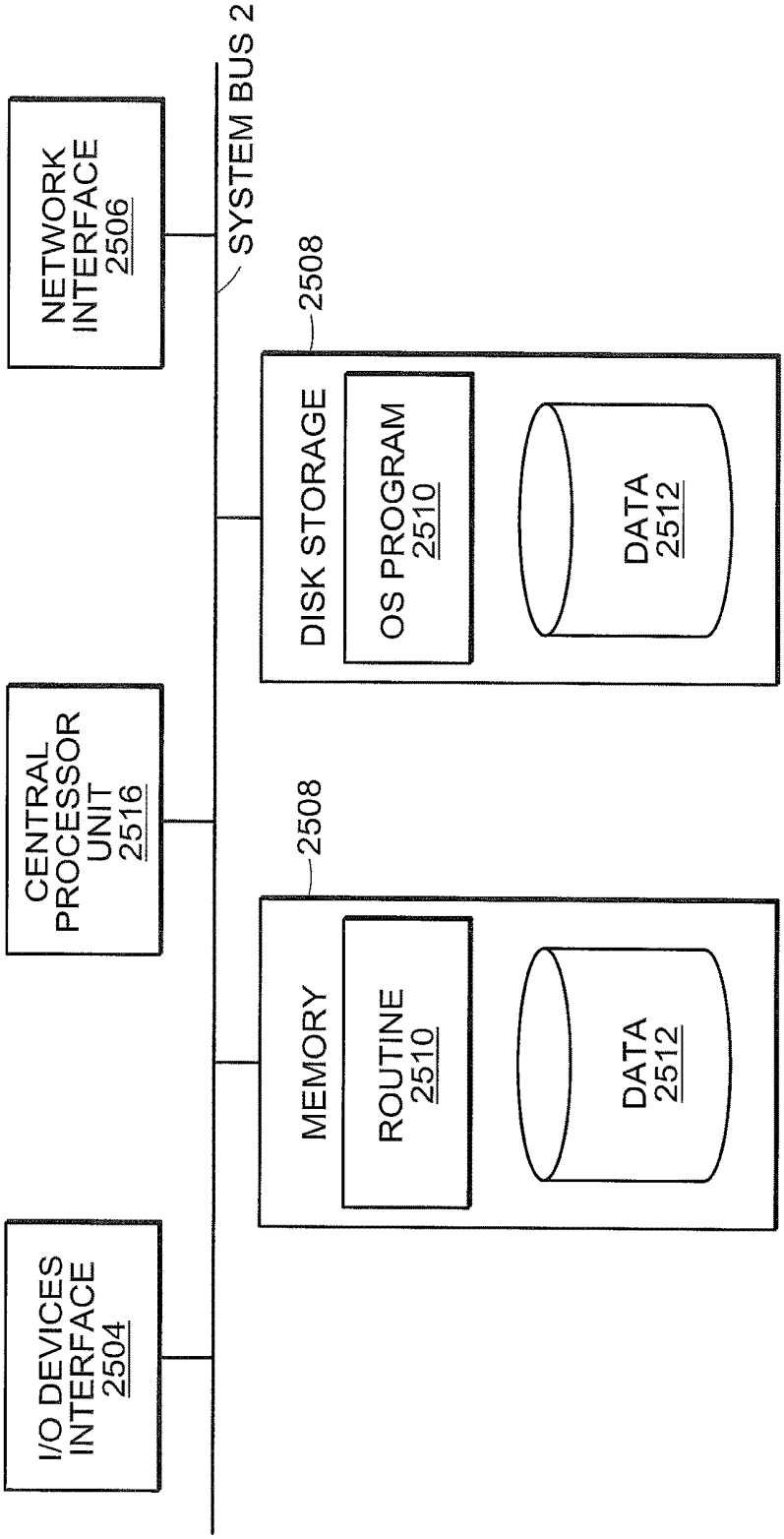
NETWORK INTERFACE 2506
SYSTEM BUS 2
2508
DISK STORAGE
OS PROGRAM 2510
DATA 2512
CENTRAL PROCESSOR UNIT 2516
2508
MEMORY
ROUTINE 2510
DATA 2512
I/O DEVICES INTERFACE 2504

FIG. 25

INTERACTIVE PATENT MAP

RELATED APPLICATIONS

This application is a continuation of U.S. application ser. No. 12/587,666, filed Oct. 8, 2009, now U.S. Pat. No. 8,335,998 which is a continuation of U.S. application ser. No. 11/648,034, filed Dec. 29, 2006, now abandoned. The entire teachings of the above applications are incorporated herein by reference.

BACKGROUND OF THE INVENTION

As the amount of prior art information continues to grow rapidly, the number of users who are new to the field of prior art searching also continues to grow. To assist users in identifying prior art, there are information gathering and retrieval tools and services. Users are likely to navigate patent and technical publication databases to search for relevant prior art. However, without assistance from a service that provides prior art search experts, a user is unlikely to find relevant prior art quickly and efficiently.

Increasingly, prior art information gathering and retrieval services are faced with a market full of users who want to be able to search for very specific prior art information, as quickly as possible, and without being burdened with false positives. Unfortunately, the currently available prior art searching technologies do not provide a user-friendly, comprehensive and versatile solution.

SUMMARY OF THE INVENTION

In today's dynamic global environment, effective intellectual property management can mean the difference between success and failure for a new product or even a company. Companies want to be able to easily evaluate and generate a comprehensive assessment of their products, technologies and intellectual property assets in the context of the prior art landscape. These users want specific prior art information that is directed to a particular type of technology and to specific companies developing that technology. Although there are prior art searching tools, they do not enable users to easily conduct effective and efficient due diligence, infringement studies, patentability searches, and freedom-to-operate studies. Thus, the current prior art information gathering and retrieval schemes are unable to provide efficiently a user with such targeted information.

The present invention relates generally to an interactive map depicting the prior art landscape. The interactive map is an effective tool for displaying and analyzing the complete intellectual property landscape within a given field. Based on content analysis of relevant patents and patent applications, the interactive map can provide a systematic review of vast quantities of data, thereby allowing the user to discern critical technology and product trends, prior art references, and the strategies of both leading and emerging competitors. In order to develop an accurate picture of the intellectual property landscape, key product features and attributes can be selected for use in analyzing the prior art documents. This yields a consistent set of information that allows further analysis. Each patent can be analyzed within the context of the prior art landscape to uncover novel features, strong claims, and business and technology trends. The maps can support visualization of major features of the landscape and provide an accurate, comprehensive picture of the intellectual property landscape. This comprehensive view can provide a foundation for creating effective corporate strategies in-tune with the realities of the intellectual property terrain.

According to an embodiment of the present invention, a computer implemented method for analyzing prior art is provided. Features of interest from a collection of prior art documents are stored. The features of interest can be stored in a database. The features of interest include technical features and dates associated with the prior art documents. The features of interest can be determined using artificial intelligence or rules based systems. An interactive map is generated that reflects the prior art features of interest. Visual indicators that represent at least a portion of the prior art documents are distributed onto the interactive map. The prior art document indicators are distributed on the interactive map based on the technical features and the dates associated with the prior art documents.

Prior art documents can be arranged on the interactive map by technical features into branches (y-axis on map) and by dates (x-axis on map). The dates used to arrange the prior art documents on the interactive map may include any of the following: a patent application filing date, a patent application publication date, a patent application priority date, and a document publication date. The prior art documents displayed on the interactive map may include patents, applications, journal articles or technical papers.

A navigation tool may be provided to enable the user to navigate the interactive map. Bibliographic information about a prior art document is displayed on the interactive map when a user hovers a mouse cursor over that prior art document's indicator represented on the map. The bibliographic information can includes any of patent number, priority date, priority country, application date, publication date, title, assignee, inventor, figure/abstract, and defined attributes.

A color schema for each country associated with prior art documents can be defined. The color schema can be used to generate a chart that compares information about the prior art documents for each country. The prior art documents for each country can be clustered on the chart by color.

BRIEF DESCRIPTION OF THE DRAWINGS

The patent or application file contains at least one drawing executed in color. Copies of this patent or patent application publication with color drawing(s) will be provided by the Office upon request and payment of the necessary fee.

The foregoing will be apparent from the following more particular description of example embodiments of the invention, as illustrated in the accompanying drawings in which like reference characters refer to the same parts throughout the different views. The drawings are not necessarily to scale, emphasis instead being placed upon illustrating embodiments of the present invention.

FIG. **1**A illustrates an example of an interactive Global Map in accordance with an embodiment of the invention.

FIG. **1**B illustrates an example of a zoomed out version of the interactive Global Map.

FIG. **1**C illustrates an example of a zoomed in version of an interactive Global Map.

FIG. **1**D illustrates an example of a navigation pop-up window.

FIG. **1**E illustrates an example of the interactive Global Map of FIG. **1**A that has been generated showing only assignee attributes.

FIG. **1**F illustrates an example of the interactive Global Map of FIG. **1**E layered over the interactive Global Map of FIG. **1**A.

FIG. **2** illustrates an example window that a user may receive upon hovering the mouse over a particular patent within the Global Map display.

FIG. **3** illustrates the type of information that can be displayed for each patent within the Global Map system.

FIG. **4** illustrates an example page that may be used to set up branch information for the patent data.

FIG. **5** illustrates an example page that enables the user to map various country codes to different color schema.

FIG. **6** illustrates an example color chooser.

FIG. **7**A illustrates an example data file.

FIG. **7**B illustrates an example layout file.

FIG. **7**C illustrates an example of an upload page.

FIG. **8** illustrates an example of a load patent and layout data page.

FIG. **9** illustrates an example of an attribute management page.

FIG. **10** illustrates an example of a page indicating that the data has been reinitialized.

FIG. **11** illustrates an example of a map generation screen.

FIG. **12** illustrates an example of a page enabling to view the global map.

FIG. **13**A illustrates an example of a small map navigation viewer.

FIG. **13**B illustrates an example of a map navigation viewer.

FIG. **14** illustrates an example of a chart generation screen.

FIG. **15** illustrates an example of a patent chart.

FIG. **16** illustrates an example of a portfolio chart generation screen.

FIG. **17** illustrates an example of a portfolio chart.

FIG. **18** is a flow diagram illustrating the process for generating a Global Map.

FIG. **19** is a diagram illustrating the computer architecture according to an embodiment of the invention.

FIG. **20** is a block diagram illustrating the database design.

FIGS. **21**A and **21**B illustrate an example of the software package file contents for map generation and static data maintenance.

FIG. **21**C is an example of the software package file contents for the web application structure.

FIG. **21**D shows an example of the file structure for the map templates software package file structure.

FIG. **21**E shows an example user interface chart software package file structure.

FIG. **22** is a block diagram illustrating an example implementation of the presentation layer architecture according to the Model, View, Control (MVC) paradigm.

FIG. **23** is a diagram illustrating the basic elements and processing flow using a template engine.

FIG. **24** illustrates a computer network or similar digital processing environment in which the present invention may be implemented.

FIG. **25** is a diagram of the internal structure of a computer in the computer system of FIG. **24**.

DETAILED DESCRIPTION OF THE INVENTION

A description of example embodiments of the invention follows.

Interactive Global Map

FIG. **1**A is an example interactive "Global Map" **100** in accordance with an embodiment of the invention. The Global Map **100** is a patent map that provides an effective and efficient tool for displaying and analyzing the complete intellectual property landscape within a given technical field.

The methodology underlying the Global Map **100** is based on content analysis, in contrast with the majority of the current generation of patent analysis tools that rely on citation analysis. Citation analysis is more than fifteen years old and is based on the relationships among patents, as evidenced by the chain of cited references. It holds that patents, which cite a relevant patent, are relevant, and that patents which are often cited are "strong patents." In contrast, a body of evidence has surfaced over the last few years that suggests cited patents may often be non-relevant, and that the frequency of citation is more a reflection of whether the terms utilized in a patent are likely to be picked up in a keyword search.

Thus, in his presentation on "The Patent Market: Positioning Your Product, Finding the Buyer" at the 40th Annual Meeting of the Licensing Executives Society, Oct. 17-21, 2004, Mr. Derek Nuhn, SVP of Semiconductor Insights reported that the patents which yielded the highest licensing revenues often had few citations, and that there appeared to be no relationship between patent value as measured by licensing and the number of cited patents. Similar findings were reported in a study carried out by Global Prior Art for the International Copper Association in 2003. A panel of industry experts selected the most important patents from among 300+ documents on the landscape. Comparing the patents selected by the experts based on content against patents which had the most citations, several trends were apparent: 1) strong patents often had few citations; and 2) patents frequently cited did not disclose distinguishing technology, but were often cited in conjunction with many continuations. The above findings were reported in a briefing with the International Copper Association at the conclusion of this research project. Others have reported similar findings.

By using a content analysis of relevant patents, patent applications and other technical documents, the Global Map **100** provides a systematic review of vast quantities of data, thereby allowing the user to discern critical technology and product trends, prior art references, and the strategies of both leading and emerging competitors. For developing an accurate picture of the intellectual property landscape, key product features and attributes are selected for use in analyzing the patents. This yields a consistent set of information that allows further analysis. Each patent is analyzed within the context of the prior art landscape to uncover novel features, strong claims, and business and technology trends. The maps support visualization of major features of the landscape and provide an accurate, comprehensive picture of the intellectual property landscape. This comprehensive view provides a foundation for creating effective corporate strategies in-tune with the realities of the intellectual property terrain. The Global Map **100** can be used for a variety of purposes, including to help target research and development funds to novel and productive research, to assess freedom-to-operate strategies, to evaluate acquisition opportunities, to identify business opportunities, and create valuable IP portfolios.

The Global Map **100** allows the user to generate various types of charts based on the patent counts and/or portfolio distribution by assignees and the technology category associated with the patent (the "branch") **102**, **104**, **106**. The charts can be generated on either patent's priority year or patent's application year.

To create a Global Map **100**, all patents and applications relevant to a specific search focus are collected, analyzed within the context of prior art, and positioned on the map. Documents are arranged by technical features, into branches **102**, **104**, **106** (y-axis on map) and by priority dates **108** (x-axis on map). In addition to patents and applications, other documents, such as journal articles or technical papers, can also be put on the map.

FIG. **1**B is an example of a zoomed out version of the interactive Global Map **100-2**. This screenshot shows how a zoomed-out version of the interactive Global Map **100-2** would appear on a user's screen, in a web browser.

FIG. **1**C is an example of a screenshot of a zoomed in version of an interactive Global Map **100-4**. This screenshot shows how a zoomed-in version of the interactive Global Map **100-4** would appear on a user's screen, in a web browser

FIG. **1**D is an example of a navigation pop-up window **140** that is used to navigate the interactive Global Map **100**. The blue-gray rectangle **142** can be dragged with a mouse cursor to the section of the map which the user is interested in viewing (or zooming in on). The user is then taken to that section on the regular interactive Global map window **100**.

It can be useful for a user to see different versions of a patent map in the context of a full landscape. For example, if the user generates an entire Global Map **100** (showing all the assignees, attributes, etc.), but then wants to see specifically where the user's corporation or its competitors are in the context of the full map **100**, the user would first generate the full map **100** as shown in FIG. **1**A. Then, the user would generate another map **100-1** shown in FIG. **1**E, detailing specific attributes of interest. In FIG. **1**E, the specific attribute of interest is assignee. In particular, the interactive Global Map **100-1** has been generated showing only assignee attributes. The user has the ability to request that a map be generated focusing on any attribute or feature of interest. FIG. **1**F illustrates an example of the interactive Global Map of FIG. **1**E layered over the interactive Global Map of FIG. **1**A. By layering **100-2** the Global Map **100-1** over Global Map **100**, and graying out Global Map **100**, the user can easily view the areas of interest. This can be even more apparent if the user toggles between Global Map **100-1** and Global Map **100-2**.

FIG. **2** illustrates the display that a user may receive upon hovering the mouse over a particular patent **200** within the Global Map **100-6** display. The map label feature includes bibliographic information **202** for the patent **200** as well as color-coded attribute boxes **204**. If the particular patent **200** is selected, then a PDF of the patent is provided.

FIG. **3** illustrates the information **300** that can be displayed for each patent within the Global Map system **100**. The information **300** includes the following: patent number **302**, priority date **304**, priority country **306**, application date **308**, publication date **310**, title **312**, assignee **314**, inventors **316**, figure/abstract **318**, and attributes one through ten **320**.

Creating the Interactive Global Map

The invention provides a web-based Internet application that allows users to generate the interactive global map (patent maps) and charts based on various patent criteria. The application could be accessible through an Internet Explorer browser (6.0+) and may be used in conjunction with Adobe SVG plug-in and Java plug-in for viewing generated maps and charts. The plug-ins are installed automatically when the user tries to view the map or chart for the first time. The user initializes the data attributes, branch, assignees, and patent information before generating maps/charts.

FIG. **4** is an example of a page **400** that may be used to set up branch information for the patent data. The user can enter in the ID **402** and BranchDesc Information **404**. The ColumnName **406** can be defaulted to the branchnumber **408**, which is a field name located in the layout file. The ID **402** refers to the label that appears on the generated map **100** and BranchDesc **404** refers to the number corresponding to the patent's branch number, which should correspond to the integer value specified for branchnumber **408** in the input data file.

FIG. **5** is an example of a page **500** that enables the user to map various country codes to different color schema. To do so, the user could type in the hexadecimal color codes **502**. The country value **504** is a two letter code which is specified in the data file.

FIG. **6** is a screenshot of an example color chooser **600**. The color chooser **600** is available by selecting tools **510** and then color chooser from the left navigation menu in FIG. **5** to help the user in obtaining the hex color code. The color chooser screen generates the hex color code for the chosen color on the screen, which can be copied or pasted into the color schema field **502** of FIG. **5**.

Next, the user enters the static data prior to generating patent maps. The static data is usually the comma-separated value text export from the database. Using the upload screen, the user uploads two files: (1) the data file (comma-separated value export) and (2) the layout file, which defines how columns are laid out. An example data file **700** is provided is shown in FIG. **7**A, and an example layout file **702** is provided in FIG. **7**B. FIG. **7**C is an example of a page **704** in which the user would select the local file in the input box and click on upload button to upload each file. The layout file **702** should not contain any blank rows at the top. The first thirteen lines specify the following patent fields:

PATENTNUMBER
PRIORITYCOUNTRY
PRIORITYDATE
APPLICATIONDATE
PUBLICATIONDATE
TITLE
INVENTORNAME
ASSIGNEEAPPLICANTNAME
ABSTRACT
SUMMARY
BRANCHNUMBER

Line **14** onwards specifies the individual attributes. Every attribute is specified in a separate line. Every attribute line entry implies both "claimed" and "taught" attributes.

Once the files are uploaded **806** onto the server, the user can initiate a data load by choosing the patent data **802** and layout files **804** on the Load Data screen, which is illustrated by FIG. **8**. First, the system will read the layout file **702** and create all mappings within the web server's database ("patentinfoattribute" table). The system then reads the comma-separated value file **700** based on the layout file and loads the data into the database ("patentinfo" table).

The screen will display the number of records inserted. Preferably, the user validates this against the expected number from the data file **700**. In case of missing records, the user can ensure that: (1) the data file **700** has the correct number of attributes as per the layout file **702**, (2) all fields are enclosed in double quotes, and (3) all fields are separated by commas.

Once the data is loaded, the user can navigate to Attribute Mappings screen **900** to validate the attribute mappings, as illustrated by FIG. **9**. The Attribute Mappings **900** screen also allows the user to make any changes. Once the layout file **702** is loaded into the system, the user can validate the attribute information using the screen **900** illustrated by FIG. **9**. This screen **900** also allows the user to make any modifications to the "attribute names," "colors, "attribute display name" on the map **100**. The map **100** can display up to ten attributes in a label.

The user navigates to the screen **1000** illustrated by FIG. **10** to synchronize the map generation server with all the loaded data prior to generating any map.

The system supports map generation based on the patent filing year, as illustrated by the screen **1100** in FIG. **11**. The user has the option to choose one or any combination of selection criteria, which is the branch **1102**, patent (filing) year **1104**, claimed **1106** and taught attributes **1108**, and assignee information **1110**. All criteria are set to "ALL" by default. Upon selecting the Generate button **1112**, the user is prompted to enter a map name. Based on the user selection criteria, the map is generated on the server with the name entered by the user. The user is taken to a new screen which indicates the successful generation of the map.

As depicted by the screen **1200** in FIG. **12**, the user can view the generated maps by choosing the map name from the drop down box **1202**. Upon clicking on "View Global Map" button **1204**, the map is loaded within a new window **1304**, as shown in FIG. **13**B. This also opens another, smaller window **1300** for easier navigation shown in FIG. **13**A. The user can hover the mouse over the desired patents to view the patent information in larger dimension. Hovering the mouse over the key icon **1306** will bring up a legend. The patents are categorized horizontally based on the various branches and filing year. The smaller navigation map **1300** provides a horizontal and a vertical slider **1302** for easy zoom in/out features. By clicking on each patent label on the main patent map, a separate window will be opened to a third party website to display the full patent text or PDF.

The screen **1400** illustrated by FIG. **14** allows generation of bar charts for patent distribution based on assignees or branch **1402**. The user selects either ASSIGNEE-PATENTINFO or BRANCHPATENTINFO from the form menu **1402**, then either Priority Date or Application Date **1404** from another menu. Patent and application years are both displayed separately, but the values selected are applied based on the selection in the year selection box.

Every patent year/application year **1502**, **1504** will have a separate chart within the tabbed display, as shown by FIG. **15**. Swing tabbed controls can be used for displaying charts based on patent year/application year.

FIG. **16** depicts the screen **1600** that allows for the generation of portfolio charts based on either branch **1602** or assignee information **1604**. The user could choose either "BRANCHPORTFOLIO" or "ASSIGNEEPORTFOLIO" from a menu **1606** on the screen. The branch **1602** and assignee information **1604** is applied based on the chart type **1606**.

Based on the chart type, every branch/assignee will be displayed in a separate tabbed display of the portfolio chart **1700**, as shown by FIG. **17**. Although a pie chart is shown in FIG. **17**, other charts and graphs may be used, such as a heat chart, line graphs, or bar graphs.

A generalized overview of the process **1800** for generating a Global Map according to an embodiment is described in FIG. **18**. At step **1802**, a user uses an external, client side database, such as Filemaker Pro, to catalog features of interest (i.e. attributes) in the set of patents being reviewed. The user notes, in the database, whether each attribute is in the teachings or claims of the patent document. The database also stores the bibliographic information of a patent (i.e. inventor, assignee, file date, publication date, number of claims, etc.). The database also tracks which branch (category) a patent is associated. Artificial intelligence or rules based systems may be used to determine and catalogue features of interest in the set of patents being reviewed.

At step **1804**, the patent information of interest is stored in the invention database **1906** (further detailed below in FIG. **19**). The database **1906** is completed once all the patent information of interest is reviewed and stored. The database **1906** may be using a server-side program, such as MySQL program which serves as the backend to collect and store data which the user uploads, and also stores the Global Maps, account information, etc. Other than the patent number, attribute and branch information, nothing is entered manually because this information can be downloaded and transferred to the database from a number of external sources (e.g. Delphion via an XML export/import).

At step **1806**, a file, such as a CSV text file, is exported from the database. The CSV text file contains the necessary fields (patent number, inventor, assignee, dates, attribute info, branch info, etc.).

At step **1808**, the data file (from the export from step **1806**) is uploaded into the database. The upload process at **1808** can be implemented using Apache commons file upload libraries, and the import functionality can be implemented using the Apache OpenCSV module.

At step **1810**, the user uploads a layout file, which is a text file where each row represents each "column" of the exported data file. The layout file helps the system map the data records into the proper fields.

At step **1812**, the system generates the global map.

System Architecture

Preferably, the invention is implemented in a software or hardware environment. One such environment is shown in FIG. **19**. In this example, a system **1900** is provided for generating an interactive Global Map **100**. The system includes an application server **1902**, a computer network **1904**, client systems **1908** for displaying the Global Map **100**, and a database **1906**, such as a MySQL application database.

FIG. **20** illustrates the database **1906** design. The database **1906** contains all of the patent information ("Patentinfo") **2002** and additional static data for map or chart generation. "Static data" includes the following: patent branch information ("BRANCHINFO") **2010**, patent attribute list ("PATENTINFOATTRLIST") **2004**, assignee to color mapping ("ASSIGNEECOLORMAPPER") **2006**, and patent country to color map ("COUNTRYCOLORMAPPER") **1008**. The user **2012** uses the database **1906** to catalog the features of interest in the set of patents being reviewed.

The User **2012**, Groups **2014**, and Permissions **2016** components enable administrators to add/maintain login accounts and access to the system **1900**. The admin would be able to create/delete/modify accounts as well as give each account permissions such as "full access, read-only access, map generation access, etc."

Each table data stored (e.g. branch **2010**, attribute **2004**, assignee **2006**, country color mappings **2008**) has a corresponding Data Access Object (DAO). The DAOs provide a common interface between the application layer and the database **1906**. FIGS. **21**A and **21**B show an example of the software package **2100-1**, **2100-2** file structure for map generation and static data maintenance including the DAOs.

Referring to FIG. **19**, using the application server **1902**, a web-based Internet application can be created that allows users to generate the interactive global map **100** and charts **1700** based on various patent criteria. The application is hosted within the application server **1902**, which delivers the content to the web **1904**. The application could be accessible through an Internet Explorer browser (6.0+) and may be used in conjunction with Adobe SVG plug-in and Java plug-in for viewing generated maps **100** and charts **1700**. SVG map actions are implemented using SVG JavaScript extensions. Portions of the Global map, such as the SVG graphic, can be exported by the user for reuse. The plug-ins are installed automatically when the user tries to view the map or chart for the first time. FIG. **21**C shows an example of the software package **2104** file contents for the web application structure used in connection with viewing and generating the patent maps **100** and charts **1700**.

The user can download the Global Maps **100** (that have been generated through the web interface) onto a CD/DVD. This feature enables users to view the maps offline. The download allows the user to download the SVG files. In addition, any HTML and JSP files (and SVG viewer plug-in, in case the user needs it) may also be downloaded so that the user can burn the batch to a CD/DVD for complete offline viewing. Any associated patents or technical documents that are identified on the Global Maps **100** are downloaded in PDF format onto a specific directory on the CD/DVD.

Referring back to FIGS. **19** and **20**, the static data (branch **2010**, attribute **2004**, assignee **2006**, country color mappings **2008**) are saved within the application server's **1902** cache for faster processing. The application server **1902** can be a Tomcat Application server.

The presentation layer is responsible for the delivery and formatting of information to the application layer. The presentation layer is developed using Java Server Pages (JSPs). In map and chart generation, the presentation layer pulls the static data (branch **2010**, attribute **2004**, assignee **2006**, country color mappings **2008**) from the application server **1902** cache before making a database **1906** request.

In an example implementation, the presentation layer architecture is split up into three layers according to the Model, View, Control (MVC) paradigm, as shown in FIG. **22**. The modules from the presentation layer **2102** are loosely coupled using the MVC design pattern. In this way, user input logic and map presentation data are separated from data access and business logic. The model code **2208** handles data and logic. The model code **2208** accesses and represents map data, and handles business operations. The view **2206** handles map output. The view code **2206** displays the global map **100** and charts **1700** to the user. The controller **2204** handles input. The controller code **2204** manipulates the model **2108** or changes the view **2206** accordingly in response to user input (the controller **2204** can also do both).

The model **2208**, view **2206**, and controller **2204** parts have specific relationships to one another. The view **2206** accesses the model **2206** to retrieve map and chart data for display. The controller **2204** receives user input and makes method invocations on the model **2208**. The controller **2204** updates the view **2206**, or selects a new view **2206**, based on the state of the model **2208** and on its own navigation rules. The model **2208** provides map and chart data to the view **2206**, but knows nothing about how the map and chart data is presented. The model **2208** also provides business services to the controller **2204**, but knows nothing about user events the controller **2204** has received.

Apache velocity templates can be used for generating the patent maps **100**. FIG. **23** is a diagram illustrating the basic elements and processing flow using a template engine **2304**. By using the velocity Java-based template engine **2304**, the Java code **2302** can be separated from the webpages, making the website more maintainable over its lifespan. The map is split into multiple templates with substitution parameters for coordinates and other display information. The substitution variables are replaced during runtime. Using the templates facilitates the development of maps **100** according to a true MVC model. FIG. **21**D shows an example of the file structure for the map templates software package **2104** file structure including VIM template scripts.

The patent charts are implemented using JFree reports. FIG. **21**E shows an example user interface chart software package file structure **2108**.

Processing Environment

FIG. **24** illustrates a computer network or similar digital processing environment in which the present invention may be implemented. Client computer(s)/devices **1908** and server computer(s) **1902** provide processing, storage, and input/output devices executing application programs and the like. Client computer(s)/devices **1908** can also be linked through communications network **1904** to other computing devices, including other client devices/processes **1908** and server computer(s) **1902**. Communications network **1904** can be part of a remote access network, a global network (e.g., the Internet), a worldwide collection of computers, Local area or Wide area networks, and gateways that currently use respective protocols (TCP/IP, Bluetooth, etc.) to communicate with one another. Other electronic device/computer network architectures are suitable.

FIG. **25** is a diagram of the internal structure of a computer (e.g., client processor/device **1908** or server computers **1902**) in the computer system of FIG. **24**. Each computer **1902**, **1908** contains system bus **2502**, where a bus is a set of hardware lines used for data transfer among the components of a computer or processing system. Bus **2502** is essentially a shared conduit that connects different elements of a computer system (e.g., processor, disk storage, memory, input/output ports, network ports, etc.) that enables the transfer of information between the elements. Attached to system bus **2502** is an Input/Output (I/O) device interface **2504** for connecting various input and output devices (e.g., keyboard, mouse, displays, printers, speakers, etc.) to the computer **1902**, **1908**. Network interface **2506** allows the computer to connect to various other devices attached to a network (e.g., network **1904** of FIG. **24**). Memory **2508** provides volatile storage for computer software instructions **2510** and data **2512** used to implement an embodiment of the present invention (e.g., web based software). Disk storage **2514** provides non-volatile storage for computer software instructions **2510** and data **2512** used to implement an embodiment of the present invention. Central processor unit **2516** is also attached to system bus **2502** and provides for the execution of computer instructions.

In one embodiment, the processor routines **2510** and data **2512** are a computer program product (generally referenced **2510**), including a computer readable medium (e.g., a removable storage medium, such as one or more DVD-ROM's, CD-ROM's, diskettes, tapes, etc.) that provides at least a portion of the software instructions for the invention system. Computer program product **2510** can be installed by any suitable software installation procedure, as is well known in the art. In another embodiment, at least a portion of the software instructions may also be downloaded over a cable, communication and/or wireless connection. In other

embodiments, the invention programs are a computer program propagated signal product embodied on a propagated signal on a propagation medium (e.g., a radio wave, an infrared wave, a laser wave, a sound wave, or an electrical wave propagated over a global network, such as the Internet, or other network(s)). Such carrier medium or signals provide at least a portion of the software instructions for the present invention routines/program **2510**.

In alternate embodiments, the propagated signal is an analog carrier wave or digital signal carried on the propagated medium. For example, the propagated signal may be a digitized signal propagated over a global network (e.g., the Internet), a telecommunications network, or other network. In one embodiment, the propagated signal is a signal that is transmitted over the propagation medium over a period of time, such as the instructions for a software application sent in packets over a network over a period of milliseconds, seconds, minutes, or longer. In another embodiment, the computer readable medium of computer program product **2510** is a propagation medium that the computer system **1908** may receive and read, such as by receiving the propagation medium and identifying a propagated signal embodied in the propagation medium, as described above for computer program propagated signal product.

Generally speaking, the term "carrier medium" or transient carrier encompasses the foregoing transient signals, propagated signals, propagated medium, storage medium and the like.

In alternate embodiments, the propagated signal is an analog carrier wave or digital signal carried on the propagated medium. For example, the propagated signal may be a digitized signal propagated over a global network (e.g., the Internet), a telecommunications network, or other network. In one embodiment, the propagated signal is a signal that is transmitted over the propagation medium over a period of time, such as the instructions for a software application sent in packets over a network over a period of milliseconds, seconds, minutes, or longer. In another embodiment, the computer readable medium of computer program product **2510** is a propagation medium that the computer system **1908** may receive and read, such as by receiving the propagation medium and identifying a propagated signal embodied in the propagation medium, as described above for computer program propagated signal product.

Generally speaking, the term "carrier medium" or transient carrier encompasses the foregoing transient signals, propagated signals, propagated medium, storage medium and the like.

While this invention has been particularly shown and described with references to preferred embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the scope of the invention encompassed by the appended claims.

For example, the present invention may be implemented in a variety of computer architectures. The computer network of FIGS. **19**, **24** and **25** are for purposes of illustration and not limitation of the present invention.

The invention can take the form of an entirely hardware embodiment, an entirely software embodiment or an embodiment containing both hardware and software elements. In a preferred embodiment, the invention is implemented in software, which includes but is not limited to firmware, resident software, microcode, etc.

Furthermore, the invention can take the form of a computer program product accessible from a computer-usable or computer-readable medium providing program code for use by or in connection with a computer or any instruction execution system. For the purposes of this description, a computer-usable or computer readable medium can be any apparatus that can contain, store, communicate, propagate, or transport the program for use by or in connection with the instruction execution system, apparatus, or device.

The medium can be an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor system (or apparatus or device) or a propagation medium. Examples of a computer-readable medium include a semiconductor or solid state memory, magnetic tape, a removable computer diskette, a random access memory (RAM), a read-only memory (ROM), a rigid magnetic disk and an optical disk. Some examples of optical disks include compact disk-read only memory (CD-ROM), compact disk-read/write (CD-R/W) and DVD.

A data processing system suitable for storing and/or executing program code will include at least one processor coupled directly or indirectly to memory elements through a system bus. The memory elements can include local memory employed during actual execution of the program code, bulk storage, and cache memories, which provide temporary storage of at least some program code in order to reduce the number of times code are retrieved from bulk storage during execution.

Input/output or I/O devices (including but not limited to keyboards, displays, pointing devices, etc.) can be coupled to the system either directly or through intervening I/O controllers.

Network adapters may also be coupled to the system to enable the data processing system to become coupled to other data processing systems or remote printers or storage devices through intervening private or public networks. Modems, cable modem and Ethernet cards are just a few of the currently available types of network adapters.

What is claimed is:

1. A data processing system configured to facilitate analysis of patent documents relevant to one or more technologies of interest, the patent documents including patents and patent applications, the system comprising:

at least one processor; and

at least one memory with computer code instructions stored thereon, the at least one processor and the at least one memory, with the computer code instructions, configured to cause the data processing system to implement:

a search handler configured to receive a selected configuration for an interactive patent map, the selected interactive patent map configuration providing an arrangement of representative patent documents for the interactive patent map based on:

(i) dates of interest from representative patent documents, the dates of interest including dates associated with respective ones of the representative patent documents;

(ii) a first technological branch that reflects a first desired technological category of interest disclosed in representative patent documents,

(iii) a second technological branch that reflects a second desired technological category of interest disclosed in representative patent documents, the representative patent documents from the first and second desired technological categories of interest being determined using a computerized content analysis derived from key features disclosed in a claims section of at least one of the respective representative patent documents;

(iv) assignee of the representative patent documents responsive to the first and second technological branches, and

(v) a distribution of visual indicators along a first axis and a second axis of the interactive patent map, the visual indicators being distributed based on technological features associated with the respective representative patent documents, such that the visual indicators representing at least a portion of the representative patent documents on the interactive map along the first axis of the interactive map based on at least one respective technological feature of the representative patent document, and along a second axis of the interactive map based on a respective date of interest of the representative patent documents;

an engine, in communication with the handler, configured to process, using a database of results, the selected interactive patent map configuration to provide a responsive interactive patent map, such that the representative patent documents responsive to the technological branches are arranged on the resulting interactive patent map according to assignee;

wherein the representative patent documents responsive to the technological branches are arranged on the resulting interactive patent map according to assignee, such that an assignee's patent documents are clustered on the resulting interactive patent map based on a technological category associated with the representative patent documents; and

wherein in response to a user hovering a mouse over a representative patent document visual indicator represented on the map, displaying bibliographic information about the respective representative patent document.

2. The data processing system as in claim **1** wherein the representative patent documents responsive to the first technological branch are arranged on the resulting interactive patent map according to assignee based on patent count, such that assignees' respective representative patent documents are clustered on the resulting interactive patent art map based on an amount of respective patent documents associated with a respective assignee, which are responsive to the first technological branch.

3. The data processing system as in claim **1** wherein the representative patent documents responsive to the first technological branch are arranged on the resulting interactive patent map according to assignee, such that an assignee's patent documents are clustered on the resulting interactive patent map based on a technological category associated with the patent documents.

4. The data processing system as in claim **1** wherein the interactive patent map is configured with a multitude of layered selected configurations, such that the first selected configuration is a first layer of the interactive patent map.

5. The data processing system as in claim **1** wherein the interactive patent map is configured to enable a user to toggle between the first and second selected configurations.

6. The data processing system as in claim **1** further includes:

the handler further configured to receive a second selected configuration for the interactive patent map, the second selected interactive patent map configuration providing an arrangement for the interactive patent map based on (i) a second technological branch that reflects a further desired technological category of interest disclosed in representative patent documents, and (ii) assignee of the representative patent documents responsive to the second technological branch; and

the engine, in communication with the handler, further configured to process, using a database of results, the selected first and second interactive patent map configurations to provide the resulting interactive patent map, such that the patent documents responsive to the first technological branch are arranged on the resulting interactive patent map according to assignee, and the patent documents responsive to the second technological branch are arranged on the resulting interactive patent map according to assignee;

wherein the first and second selected configurations are configured as respective first and second layers of the interactive patent map.

7. The data processing system as in claim **6** wherein the dates associated with the patent documents include any of the following: a patent application filing date, a patent application publication date, a patent application priority date, and a document publication date.

8. The data processing system as in claim **4** wherein the distributed visual indicators for the respective patent documents are responsive to the second technological branch along the first axis and the second axis of the interactive patent map including:

arranging the visual indicators based on technological features associated with the respective representative patent documents to be responsive to the second technological branch according to the y-axis on the interactive patent map; and

arranging the visual indicators based on the dates according to the x-axis on the interactive patent map.

9. The data processing system as in claim **1** wherein the database of results is interfaced with using a map generation server having preloaded patent map generation data including attributes of a multitude of patent documents.

10. The data processing system as in claim **9** wherein the preloaded patent map generation data includes patent related contributions from a multitude of users.

11. The data processing system as in claim **1** wherein at least one web browser plug-in is automatically transmitted when a user attempts to view the interactive patent map for the first time.

12. The data processing system as in claim **1** wherein the interactive patent map is configured to be exported for offline use.

13. The data processing system as in claim **1** wherein the interactive patent map is exported as a PDF file.

14. The data processing system as in claim **1** wherein the patent documents responsive to the first technological branch are clustered on the resulting interactive patent map according to assignee by color.

15. The data processing system as in claim **1** wherein the engine is configured to dynamically update the selected first interactive patent map configuration to reflect a new selected interactive patent map configuration.

16. The data processing system as in claim **6** wherein the first technological branch includes one or more of technology, manufacturing, or product features.

17. The data processing system as in claim **16** wherein the first technological branch further includes key features derived from a claims section of at least one of the patent documents.

18. The data processing system as in claim **1** wherein the engine is configured to be responsive to requests to display bibliographic information concerning the respective patent documents responsive to the first technological branch.

19. The data processing system as in claim 18 wherein the request to display bibliographic information is a detected mouse event; and

wherein the engine is configured to respond to the mouse event by direct display of bibliographic information about a patent document.

20. The data processing system as in claim 19 wherein the detected mouse event includes mouse click and hovering events; and

the event handler configured to respond to the mouse click and hovering events by showing the bibliographic information including one or more of: a patent number, priority date, priority country, application date, publication date, title, assignee, inventor, figure, abstract, and defined attributes.

21. The data processing system as in claim 20 wherein the interactive patent map provides a visual representation of the patent landscape for use in connection with a due diligence investigation, infringement study, patentability search, or freedom-to-operate study.

22. The data processing system as in claim 1 wherein the interactive patent map is generated from a layout file and a file CSV file using the database of results.

23. The data processing system as in claim 1 wherein a user initializes data attributes, branch, assignees, and patent information before generating maps/charts.

24. A data processing system configured to facilitate analysis of patent documents relevant to one or more technologies of interest, the patent documents including patents and patent applications, the system comprising:

at least one processor; and

at least one memory with computer code instructions stored thereon, the at least one processor and the at least one memory, with the computer code instructions, configured to cause the data processing system to implement:

a handler configured to receive a first selected configuration for an interactive patent map, the first selected interactive patent map configuration providing an arrangement of representative patent documents for of the interactive patent map based on:

(i) dates of interest from patent documents, the dates of interest including dates associated with respective ones of the representative patent documents;

(ii) a first technological branch that reflects a desired technological category of interest disclosed in representative patent documents, and (iii) assignee of the representative patent documents responsive to the first technological branch;

the handler configured to receive a second selected configuration for the interactive patent map, the second selected interactive patent map configuration providing an arrangement for the interactive patent map based on (i) a second technological branch that reflects a further desired technological category of interest disclosed in representative patent documents, (ii) assignee of the representative patent documents responsive to the second technological branch, and including a distribution of visual indicators along a first axis and a second axis of the interactive patent map, the visual indicators being distributed based on technological features associated with the respective patent documents;

the representative patent documents from the first and second desired technological categories of interest being determined using a computerized content analysis derived from key features disclosed in a claims section of at least one of the respective representative patent documents, such that the visual indicators representing at least a portion of the representative patent documents on the interactive map along the first axis of the interactive map based on at least one respective technological feature of the representative patent document, and along a second axis of the interactive map based on a respective date of interest of the representative patent documents;

an engine, in communication with the handler, configured to process, using a database of results, the selected first and second interactive patent map configurations to provide a resulting interactive patent map, such that the patent documents responsive to the first technological branch are arranged on the resulting interactive patent map according to assignee, and the patent documents responsive to the second technological branch are arranged on the resulting interactive patent map according to assignee;

wherein the patent documents responsive to the technological branches are arranged on the resulting interactive patent map according to assignee, such that an assignee's patent documents are clustered on the resulting interactive patent map based on a technological category associated with the representative patent documents; and

wherein in response to a user hovering a mouse over a representative patent document visual indicator represented on the map, displaying a bibliographic information about the respective representative patent document.

25. A method configured to facilitate analysis of patent documents relevant to one or more technologies of interest, the patent documents including patents and patent applications, the system comprising:

receiving a selected configuration for an interactive patent map, the selected interactive patent map configuration providing an arrangement of representative patent documents for the interactive patent map based on:

(i) dates of interest from patent documents, the dates of interest including dates associated with respective ones of the representative patent documents,

(ii) a first technological branch that reflects a first desired technological category of interest disclosed in representative patent documents,

(iii) a second technological branch that reflects a second desired technological category of interest disclosed in representative patent documents,

(iv) assignee of the representative patent documents responsive to the first and second technological branches; and including a distribution of visual indicators along a first axis and a second axis of the interactive patent map, the visual indicators being distributed based on technological features associated with the respective patent documents, the representative patent documents from the first and second desired technological categories of interest being determined using a computerized content analysis derived from key features disclosed in a claims section of at least one of the respective representative patent documents, such that the visual indicators representing at least a portion of the representative patent documents on the interactive map along the first axis of the interactive map based on at least one respective technological feature of the representative patent document, and along a second axis of the interactive map based on a respective date of interest of the representative patent documents;

processing, using a database of results, the selected interactive patent map configuration to provide a responsive interactive patent map, such that the patent documents responsive to the technological branches are arranged on the resulting interactive patent map according to assignee;

wherein the patent art documents responsive to the technological branches are arranged on the resulting interactive patent map according to assignee, such that an assignee's patent documents are clustered on the resulting interactive patent map based on a technological category associated with the patent documents; and

wherein in response to a user hovering a mouse over a representative patent document visual indicator represented on the map, displaying bibliographic information about the respective representative patent document.

26. A computer program product, stored on a non-transitory computer readable medium, configured to facilitate analysis of representative patent documents relevant to one or more technologies of interest, the patent documents including patents and patent applications, the computer program product having computer readable instructions, which cause at least one computer processor, to:

facilitate user selection for a configuration for an interactive patent map, the selected interactive patent map configuration providing an arrangement for the interactive patent map based on

(i) dates of interest from patent documents, the dates of interest including dates associated with respective ones of the representative patent documents,

(ii) a first technological branch that reflects a desired technological category of interest disclosed in representative patent documents,

(iii) a second technological branch that reflects a second desired technological category of interest disclosed in representative patent documents,

(iv) assignee of the representative patent documents responsive to the first and second technological branches; and including a distribution of visual indicators along a first axis and a second axis of the interactive patent map, the visual indicators being distributed based on technological features associated with the respective patent documents, the representative patent documents from the first and second desired technological categories of interest being determined using a computerized content analysis derived from key features disclosed in a claims section of at least one of the respective representative patent documents, such that the visual indicators representing at least a portion of the representative patent documents on the interactive map along the first axis of the interactive map based on at least one respective technological feature of the representative patent document, and along a second axis of the interactive map based on a respective date of interest of the representative patent documents;

process, using a database of results, the selected interactive patent map configuration to provide a responsive interactive patent map, such that the patent documents responsive to the technological branches are arranged on the resulting interactive patent map according to assignee, where an assignee's patent documents are clustered on the resulting interactive patent map based on a technological category associated with the representative patent documents; and

wherein in response to a user hovering a mouse over a representative patent document visual indicator represented on the map, displaying bibliographic information about the respective representative patent document.

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